

Numerical Methods

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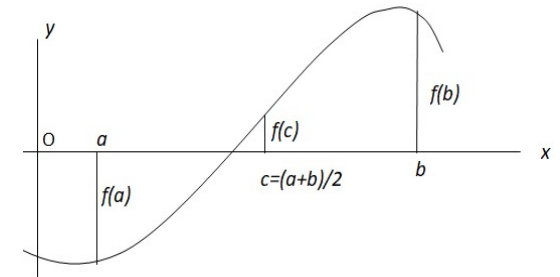
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Solution of equations

Bisection Method. The Bisection

Method is applicable for numerically solving the equation $f(x) = 0$ for the real variable x , where $f(x)$ is a continuous function defined on an interval $[a, b]$ and where $f(a)$ and $f(b)$ have opposite signs. In this case, by the intermediate value theorem, the continuous function $f(x)$ must have at least one root in the interval (a, b) .



At each step the method divides the interval $[a, b]$ in two equal subintervals $[a, c]$ and $[c, b]$ by computing the midpoint $c = \frac{a+b}{2}$ of the interval $[a, b]$. Unless c is itself a root (which is very unlikely, but possible) there are now only two possibilities: either $f(a)$ and $f(c)$ have opposite signs and there is a root between a and c , or $f(c)$ and $f(b)$ have opposite signs and there is a root between c and b . Explicitly, if $f(a)$ and $f(c)$ have opposite signs, then the method sets c as the new value for b , and if $f(b)$ and $f(c)$ have opposite signs then the method sets c as the new a . (If $f(c) = 0$ then c may be taken as the solution and the process stops.) In both cases, the new $f(a)$ and $f(b)$ have opposite signs, so the method is applicable to this smaller interval. In this way an interval that contains a zero of $f(x)$ is reduced in width by 50% at each step. The process is continued until the interval is sufficiently small, i.e., $f(c)$ is near to zero.

Ex1. Find a real root of the equation $f(x) = x^3 - x - 1 = 0$.

Solution. We get

$$f(1) = 1^3 - 1 - 1 = -1 \quad (-ve)$$

$$f(2) = 2^3 - 2 - 1 = 5 \quad (+ve)$$

Hence there is a root between 1 & 2 so we take $x_0 = \frac{1+2}{2} = 1.5$

Now, $f(x_0) = f(1.5) = 1.5^3 - 1.5 - 1 = 0.875 \quad (+ve)$

Hence there is a root between 1 & x_0 . So we take $x_1 = \frac{1+x_0}{2} = \frac{1+1.5}{2} = 1.25$

We find, $f(x_1) = f(1.25) = (1.25)^3 - 1.25 - 1 = -0.296875 \quad (-ve)$

Hence the root lies between x_0 & x_1 . So we take $x_2 = \frac{x_0+x_1}{2} = \frac{1.5+1.25}{2} = 1.375$

We obtain, $f(x_2) = f(1.375) = (1.375)^3 - 1.375 - 1 = 0.2246 \quad (+ve)$

Hence the root lies between x_1 & x_2 . So we take $x_3 = \frac{x_1+x_2}{2} = \frac{1.25+1.375}{2} = 1.3125$

We obtain, $f(x_3) = f(1.3125) = (1.3125)^3 - 1.3125 - 1 = -0.051 \quad (-ve)$

So we take $x_4 = \frac{x_2+x_3}{2} = 1.34375$

Now, $f(x_4) = 0.0826 \quad (+ve)$

So let $x_5 = \frac{x_3 + x_4}{2} = 1.328125$

Now, $f(x_5) = 0.0145$, which is near to zero.

Hence the appropriate root of the given equation is 1.328125.

Ex 2. Use bisection method determine the root of $f(x) = e^{-x} - x = 0$

Solution. we get, $f(0) = 1 - 0 = 1$ (+ve)

$$f(1) = -0.63 \quad (-ve)$$

Hence the root lies between 0 & 1 so we take $x_0 = \frac{0+1}{2} = 0.5$

Now, $f(x_0) = f(0.5) = 0.1065$ (+ve)

Hence the root lies between 1 & x_0 . So we take $x_1 = \frac{1+x_0}{2} = \frac{1+0.5}{2} = 0.75$

We find, $f(x_1) = f(0.75) = -0.278$ (-ve)

Hence the root lies between x_0 & x_1 . So we take $x_2 = \frac{x_0+x_1}{2} = \frac{0.5+0.75}{2} = 0.625$

We obtain, $f(x_2) = f(0.625) = -0.088$ (-ve)

Hence the root lies between x_0 & x_2 . So we take $x_3 = \frac{x_0+x_2}{2} = \frac{0.5+0.625}{2} = 0.5625$

We obtain, $f(x_3) = f(0.5625) = 0.0073$, which is near to zero.

Hence the appropriate root of the given equation is 0.5625.

Ex 3. Perform five interaction of the bisection method to determine the smallest positive real root of $f(x) = x^3 - 5x + 1 = 0$

Solution. we get $f(0) = 1$ (+ve)

$$f(1) = -3 \quad (-ve)$$

Hence the root lies between 0 & 1, So we take $x_0 = \frac{0+1}{2} = 0.5$

Now, $f(x_0) = f(0.5) = -0.1375$ (-ve)

Hence the root lies between 0 & x_0 . So we take $x_1 = \frac{0+x_0}{2} = \frac{0+0.5}{2} = 0.25$

We find, $f(x_1) = f(0.25) = -0.234$ (-ve)

Hence the root lies between 0 & x_1 . So we take $x_2 = \frac{0+x_1}{2} = \frac{0+0.25}{2} = 0.125$

We obtain, $f(x_2) = f(0.125) = 0.377$ (+ve)

Hence the root lies between x_1 & x_2 . So we take $x_3 = \frac{x_1+x_2}{2} = \frac{0.25+0.125}{2} = 0.1875$

We obtain, $f(x_3) = f(0.1875) = 0.069$ (+ve)

Hence the root lies between x_1 & x_3 . So we take $x_4 = \frac{x_1+x_3}{2} = \frac{0.25+0.1875}{2} = 0.21875$

We obtain, $f(x_4) = f(0.21875) = -0.08$ (-ve)

Hence the root lies between x_3 & x_4 . So we take $x_5 = \frac{x_3+x_4}{2} = \frac{0.1875+0.21875}{2} = 0.203125$

We obtain, $f(x_5) = f(0.203125) = -0.007$, which is near to zero.

Hence the smallest positive real root of the given equation is 0.20325

Ex 4. Use bisection method find x_3 of the equation $\sqrt{x} - \cos x = 0$ on $[0,1]$

Let, $f(x) = \sqrt{x} - \cos x$

Solution. we get $f(0) = -1$ (-ve)

$$f(1) = 0.45970 \quad (+ve)$$

Hence the root lies between 0 & 1. So we take $x_1 = \frac{0+1}{2} = 0.5$

Now, $f(x_1) = f(0.5) = -0.17048$ (-ve)

Hence the root lies between 1 & x_1 . So we take $x_2 = \frac{1+x_1}{2} = \frac{1+0.5}{2} = 0.75$

We find, $f(x_2) = f(0.75) = 0.13434$ (+ve)

Hence the root lies between x_1 & x_2 so we take $x_3 = \frac{x_1+x_2}{2} = \frac{0.5+0.75}{2} = 0.625$

Hence $x_3 = 0.625$

Ex 5. Use the bisection algorithm to find an approximation to $\sqrt{3}$.

Solution. Let, $x = \sqrt{3}$

$$\text{or, } x^2 = 3$$

$$\text{or, } x^2 - 3 = 0$$

Therefore $\sqrt{3}$ is a root of $f(x) = x^2 - 3 = 0$

We get, $f(1) = 1^2 - 3 = -2$ (-ve)

$$f(2) = 2^2 - 3 = 1 \quad (+ve)$$

So, there is a root between 1 and 2. Hence we take $x_1 = \frac{1+2}{2} = 1.5$

Now, $f(x_1) = f(1.5) = -0.75$ (-ve)

So we obtain $x_2 = \frac{2+x_1}{2} = \frac{2+1.5}{2} = 1.75$

We find, $f(x_2) = f(1.75) = 0.0625$ (+ve)

Hence the root lies between 1.5 & 1.75 so we take $x_3 = \frac{1.5+1.75}{2} = 1.625$

We obtain, $f(x_3) = f(1.625) = -0.3593$ (-ve)

Hence the root lies between x_2 & x_3 . So we take $x_4 = \frac{x_2+x_3}{2} = \frac{1.75+1.625}{2} = 1.6875$

We obtain, $f(x_4) = f(1.6875) = -0.15234$ (-ve)

Hence the root lies between x_2 & x_4 . So we take $x_5 = \frac{x_2+x_4}{2} = \frac{1.75+1.6875}{2} = 1.71875$

We obtain, $f(x_5) = f(1.71875) = -0.045899$ (-ve)

Hence the root lies between x_2 & x_5 so we take $x_6 = \frac{x_2+x_5}{2} = \frac{1.75+1.71875}{2} = 1.734375$

We obtain, $f(x_6) = f(1.734375) = 0.008$, which is near to zero.

Hence 1.734375 is a root of $x^2 - 3 = 0$

$$\therefore \sqrt{3} = 1.734375$$

Method of False Position. The False Position Method is applicable for numerically solving the equation $f(x) = 0$ for the real variable x , where $f(x)$ is a continuous function defined on an interval $[a, b]$ and where $f(a)$ and $f(b)$ have opposite signs. In this case, by the intermediate value theorem, the continuous function $f(x)$ must have at least one zero in the interval (a, b) .

At each step the method replaces the curve $f(x)$ by the straight line joining the points $(a, f(a))$ and $(b, f(b))$. The equation of this line is

$$\frac{x-a}{a-b} = \frac{y-f(a)}{f(a)-f(b)} \dots (1)$$

If this line intersects the x axis at c (here $y = 0$), then we get the value of

c by solving (1) for x with $y = 0$. Therefore $c = a - \frac{a-b}{f(a)-f(b)} f(a)$. Unless c is itself a root (which is very unlikely, but possible) there are now only two possibilities: either $f(a)$ and $f(c)$ have opposite signs and there is a root between a and c , or $f(c)$ and $f(b)$ have opposite signs and there is a root between c and b . Explicitly, if $f(a)$ and $f(c)$ have opposite signs, then the method sets c as the new value for b , and if $f(b)$ and $f(c)$ have opposite signs then the method sets c as the new a . (If $f(c) = 0$ then c may be taken as the solution and the process stops.) In both cases, the new $f(a)$ and $f(b)$ have opposite signs, so the method is applicable to this smaller interval. The process is continued until the interval is sufficiently small, i.e., $f(c)$ is near to zero.

Ex 6. Find a real root of $f(x) = x^3 - 2x - 5 = 0$ by method of false position.

Solution. we have

$$f(2) = -1 \quad (-ve) \quad [\text{Here } x_0 = 2]$$

$$f(3) = 16 \quad (+ve) \quad [\text{Here } x_1 = 3]$$

$$\text{So, } x_2 = 2 - \frac{f(2)}{f(3)-f(2)} (3-2) = 2 - \frac{(-1)}{16-(-1)} = 2 + \frac{1}{17} = 2.0588$$

$$\text{Now, } f(x_2) = f(2.0588) = -0.386 \quad (-ve)$$

and hence the root lies between 2.0588 and 3.

$$\text{So, } x_3 = 2.0588 - \frac{f(2.0588)}{f(3)-f(2.0588)} (3-2.0588) = 2.0588 + \frac{(.386)}{.6+0.386} (3-2.0588) = 2.0812$$

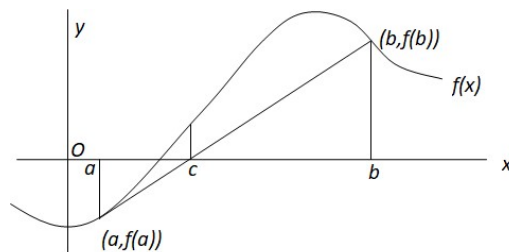
$$\text{We find, } f(x_3) = f(2.0812) = -0.1479 \quad (-ve)$$

Hence the root lies between 2.0812 and 3

$$\text{So, } x_4 = 2.0812 - \frac{f(2.0812)}{f(3)-f(2.0812)} (3-2.0812) = 2.0812 + \frac{(.1479)}{16+0.1479} (3-2.0812) = 2.0896$$

$$\text{Now, } f(x_4) = f(2.0896) = -0.0551 \quad (-ve)$$

hence the root lies between 2.0896 and 3



$$\text{So, } x_5 = 2.0896 - \frac{f(2.0896)}{f(3)-f(2.0896)} (3-2.0896) = 2.0588 + \frac{(.0551)}{16+.0551} (3-2.0896) = 2.0927$$

Hence the root of the given equation is 2.0927.

Ex 7. Solve $x^3 - x^2 - 1 = 0$ by the method of false position.

Solution. Let, $f(x) = x^3 - x^2 - 1$

$$\text{So, } f(1) = -1 < 0$$

$$f(2) = 3 > 0$$

Hence the root lies between 1 and 2.

$$\text{So, } x_2 = 1 - \frac{f(1)}{f(2)-f(1)} (2-1) = 1 + \frac{1}{3+1} \times 1 = 1.25$$

$$\text{We find, } f(x_2) = f(1.25) = -0.60938 < 0$$

Hence the root lies between 1.25 and 2.

$$\text{So, } x_3 = 1.25 - \frac{f(1.25)}{f(2)-f(1.25)} (2-1.25) = 1.25 + \frac{0.60938}{3+0.60938} (.75) = 1.3766$$

$$\text{Now, } f(x_3) = f(1.3766) = -0.28633 < 0.$$

Hence the root lies between 1.3766 and 2.

$$\text{So, } x_4 = 1.3766 - \frac{f(1.3766)}{f(2)-f(1.3766)} (2-1.3766) = 1.3766 + \frac{0.28633}{3+0.28633} (.6234) = 1.4304$$

$$\text{Now, } f(x_4) = f(1.4304) = -0.11938 < 0$$

Hence the root lies between 1.4304 and 2.

$$\text{So, } x_5 = 1.4304 - \frac{f(1.4304)}{f(2)-f(1.4304)} (2-1.4304) = 1.4304 + \frac{0.11938}{3+0.11938} (.5696) = 1.4522$$

$$\text{Now, } f(x_5) = f(1.4522) = -0.046362 < 0$$

hence the root lies between 1.4522 and 2.

$$\text{So, } x_6 = 1.4522 - \frac{f(1.4522)}{f(2)-f(1.4522)} (2-1.4522) = 1.4522 + \frac{0.046362}{3+0.046362} (.5478) = 1.4595$$

$$\text{Now, } f(x_6) = f(1.4595) = -0.0212 < 0$$

hence the root lies between 1.4595 and 2.

$$\text{So, } x_7 = 1.4595 - \frac{f(1.4595)}{f(2)-f(1.4595)} (2-1.4595) = 1.4595 + \frac{0.0212}{3+0.0212} (.5405) = 1.4633$$

$$\text{Now, } f(x_7) = f(1.4633) = -0.00796, \text{ which is near to zero.}$$

Hence the root is, $x_7 = 1.4633$

Iteration (Fixed Point Iteration) Method. The iteration method in numerical analysis is used to find an approximate solution of algebraic and transcendental equations. The method uses the concept of a fixed point in a repeated manner to compute the solution of the given equation. A fixed point is a point in the domain of a function ϕ such that $\phi(x) = x$. In the iteration method, the given function is algebraically converted in the form of $\phi(x) = x$.

Suppose we have an equation $f(x) = 0$, for which we have to find the solution. At first we have to choose the initial value x_0 for the iterative method where x_0 is some initial

guess called fixed point iterative scheme. One way to choose x_o is to find the values $x = a$ and $x = b$ for which $f(a)$ and $f(b)$ are opposite signs. By narrowing down the selection of a and b , take x_o as the average of a and b . Now the equation can be expressed as $x = \varphi(x)$. Choose $\varphi(x)$ such that $|\varphi'(x)| < 1$ at $x = x_o$. If there are more than one possibility of $\varphi(x)$, choose the $\varphi(x)$ which has the minimum value of $|\varphi'(x)|$ at $x = x_o$. Then the iterative method is applied by successive approximations given by $x_n = \varphi(x_{n-1})$, that is, $x_1 = \varphi(x_o)$, $x_2 = \varphi(x_1)$ and so on. If $f(x)$ is a continuous function, we get a sequence $\{x_n\}$ which converges to a point I_o . This I_o is the approximate solution of the given equation.

Ex 8. Find a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ by iteration method.

Solution. we get

$$f(0) = -1 \quad (-ve)$$

$$f(1) = 1 \quad (+ve)$$

So, a root lies between 0 and 1. Therefore we can take $x_o = 0.5$

To find this root, we put the equation in the form $x = \varphi(x)$.

$$\text{So } x^3 + x^2 - 1 = 0$$

$$\text{or, } x = \frac{1}{\sqrt{1+x}}$$

$$\text{So that } \varphi(x) = \frac{1}{\sqrt{1+x}} \text{ and } \varphi'(x) = \frac{1}{2(1+x)^{3/2}}$$

$$\text{We have } |\varphi'(x)| < 1 \quad \text{for } x = x_o = 0.5$$

Hence the iteration method can be applied.

We get,

$$x_1 = \varphi(x_o) = \varphi(.5) = 0.81649$$

$$x_2 = \varphi(x_1) = \varphi(.81649) = .74196$$

$$x_3 = \varphi(x_2) = \varphi(.74196) = .75767$$

$$x_4 = \varphi(x_3) = \varphi(.75767) = .75427$$

$$x_5 = \varphi(x_4) = \varphi(.75427) = .75500$$

$$x_6 = \varphi(x_5) = \varphi(.75500) = .75485$$

$$x_7 = \varphi(x_6) = \varphi(.75485) = .75488$$

Hence the approximate value of the root is 0.75488.

Ex 9. Find the root of the equation $2x = \cos x + 3$ correct to three decimal places by using iteration method.

Solution. The given equation can be put in the form, $x = \frac{1}{2}(\cos x + 3)$

$$\text{So that } \varphi(x) = \frac{1}{2}(\cos x + 3)$$

$$\text{And } \varphi'(x) = \frac{1}{2}(-\sin x)$$

$$\text{We have } |\varphi'(x)| = \left| \frac{\sin x}{2} \right| < 1 \quad \text{for any values of } x.$$

Hence the iteration method can be applied.

Take, $x_o = 1$

We get,

$$x_1 = \varphi(x_o) = \varphi(1) = 1.7701$$

$$x_2 = \varphi(x_1) = \varphi(1.7701) = 1.40098$$

$$x_3 = \varphi(x_2) = \varphi(1.40098) = 1.5845$$

$$x_4 = \varphi(x_3) = \varphi(1.5845) = 1.4931$$

$$x_5 = \varphi(x_4) = \varphi(1.4931) = 1.5388$$

$$x_6 = \varphi(x_5) = \varphi(1.5388) = 1.5160$$

$$x_7 = \varphi(x_6) = \varphi(1.5160) = 1.5274$$

$$x_8 = \varphi(x_7) = \varphi(1.5274) = 1.5217$$

$$x_9 = \varphi(x_8) = \varphi(1.5217) = 1.5245$$

$$x_{10} = \varphi(x_9) = \varphi(1.5245) = 1.5231$$

Hence the approximate value of the root is 1.5231.

Ex 10. Find a real root of $\sin x = 10(x - 1)$ correct to four significant figures by using iteration method.

Solution. The equation can be written in the form,

$$10x - 10 = \sin x$$

or,

$$10x = 10 + \sin x$$

or,

$$x = \frac{1}{10} \sin x + 1$$

So that,

$$\varphi(x) = \frac{1}{10} \sin x + 1$$

And

$$\varphi'(x) = \frac{-\cos x}{10}$$

We have $|\varphi'(x)| < 1$ for any values of x . Hence the iteration method can be applied.

We get,

$$x_n = \varphi(x_{n-1})$$

Choose, $x_o = 1$, we get

$$x_1 = \varphi(x_o) = \varphi(1) = 1.0841$$

$$x_2 = \varphi(x_1) = \varphi(1.0841) = 1.08834$$

$$x_3 = \varphi(x_2) = \varphi(1.08834) = 1.08859$$

$$x_4 = \varphi(x_3) = \varphi(1.08859) = 1.08860$$

$$x_5 = \varphi(x_4) = \varphi(1.08860) = 1.08860$$

Hence the approximate value of the root is 1.089.

Ex 11. Use the iteration method to find a real root of the following equations.

$$(a) \cos x = 3x - 1$$

$$(e) e^{-x} = 10x$$

$$(b) x = \frac{1}{(x+1)^2}$$

$$(f) x = \operatorname{cosec} x$$

$$(c) x = (5 - x)^{\frac{1}{3}}$$

$$(g) \sin^2 x = x^2 - 1$$

(d) $\sin x = 10(x - 1)$

(h) $e^x = \cot x$

Newton-Raphson Method. The Newton-Raphson method (also known as Newton's method) is a way to quickly find a good approximation for the root of a real-valued function $f(x) = 0$. Let x_0 be an approximate root of $f(x) = 0$ and let $x_1 = x_0 + h$ be the correct root so that

$$f(x_1) = 0$$

That is,

$$f(x_0 + h) = 0$$

Now expanding $f(x_0 + h)$ by Taylor's series, we obtain,

$$f(x_0) + h f'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots = 0$$

Neglecting the second and higher order derivatives, we have

$$f(x_0) + h f'(x_0) = 0$$

or,

$$h = -\frac{f(x_0)}{f'(x_0)}$$

A better approximation than x_0 is therefore given by x_1 when,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Successive approximations are given by $x_2, x_3, \dots, x_{n+1}$

Where,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This is Newton – Raphson formula.

Ex 12. The equation $x^3 + 2x^2 - 5x - 1 = 0$ has a root near to 1.4. Use the Newton-Raphson method to find the root to four significant figures.

Solution. We have,

$$f(x) = x^3 + 2x^2 - 5x - 1$$

$$\text{Therefore } f'(x) = 3x^2 + 4x - 5$$

Given that, $x_0 = 1.4$

$$\text{Now, } f(x_0) = -1.336$$

$$f'(x_0) = 1.6162$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.4 - \frac{-1.336}{1.6162} = 1.6062$$

$$\text{Now, } f(x_1) = .2723$$

$$f'(x_1) = 9.1641$$

$$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.6062 - \frac{.2723}{9.1641} = 1.5765$$

$$\text{Now, } f(x_2) = .0059$$

$$f'(x_2) = 8.7615$$

$$\therefore x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.5765 - \frac{.0059}{8.7615} = 1.5757$$

Now, $f(x_3) = .000003$ which is near to zero.

Hence the approximate value of the root is 1.5757

Ex 13. Use Newton's method to find a root of the equation $x^3 - 3x - 5 = 0$

Solution. We have, $f(x) = x^3 - 3x - 5 = 0$

$$f(2) = 2^3 - 3 \cdot 2 - 5 = -3$$

$$f(3) = 3^3 - 3 \cdot 3 - 5 = 13$$

So, there is a root between 2 & 3.

$\therefore$ We choose $x_0 = 2$

Now,

$$f'(x) = 3x^2 - 3$$

$\therefore$

$$f(x_0) = f(2) = -3$$

And

$$f'(x_0) = f'(2) = 9$$

$\therefore$,

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{-3}{9} = 2.3333$$

Now,

$$f(x_1) = 0.7037$$

$$f'(x_1) = 13.333$$

$\therefore$,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.3333 - \frac{0.7037}{13.333} = 2.2805$$

Now,

$$f(x_2) = f(2.2805) = 0.01935$$

$$f'(x_2) = f'(2.2805) = 12.6028$$

$\therefore$,

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.2805 - \frac{0.01935}{12.6028} = 2.2790$$

Now, $f(x_3) = f(2.2790) = 0.000016$, which is near to zero.

Hence the approximate value of the root is 2.2790.

Ex 14. Find a root of the equation $x \sin x + \cos x = 0$

Solution. We have, $f(x) = x \sin x + \cos x$

$$f'(x) = x \cos x$$

Choose, $x_0 = \pi = 3.1416$

$\therefore$

$$f(x_0) = -1$$

$$f'(x_0) = -3.1416$$

$\therefore$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3.1416 - \frac{-1}{-3.1416} = 2.8233$$

Now,

$$f(x_1) = -0.06618$$

$$f'(x_1) = -2.6816$$

$\therefore$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.8233 - \frac{-0.06618}{-2.6816} = 2.7985$$

Now, $f(x_2) = -0.00056$, which is near to zero.

Hence the approximate value of the root is 2.7985.

Ex 15. Solve the following equations by Newton's method

(i) $x^3 - 2x - 5 = 0$

(ii) $x^5 + 5x + 1 = 0$

(iii) $x^3 - 5x + 3 = 0$

(iv) $\sin x = 1 - x$

(v) $\tan x = 4x$.

Interpolation with equal intervals

Newton's forward difference interpolation formula. Given the set of $(n + 1)$ values viz. $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, \dots, (x_n, y_n)$, of x and y , it is required to find $y_n(x)$, a polynomial of the n th degree such that y and $y_n(x)$ agree at the tabulated points. Let the values of x be equidistant. i.e., Let $x_i = x_0 + ih$ for $i = 1, 2, 3, \dots, \dots, n$

Since $y(x)$ is a polynomial of the n th degree, it may be written as

$$y_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2) + \dots + a_n(x - x_0)(x - x_1)(x - x_2) \dots (x - x_{n-1}) \dots (1)$$

Imposing now the condition that y and $y_n(x)$ should agree at the set of tabulated values.

When $x = x_0$ then $y_0 = a_0$

When $x = x_1$ then $y_1 = a_0 + a_1(x - x_0)$

$$\Rightarrow y_1 = a_0 + a_1 h$$

$$\Rightarrow a_1 h = y_1 - a_0$$

$$\Rightarrow a_1 h = y_1 - y_0$$

$$\Rightarrow a_1 h = \Delta y_0$$

$$\therefore a_1 = \frac{\Delta y_0}{h}$$

When $x = x_2$ then $y_2 = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$

$$\Rightarrow y_2 = y_0 + a_1 2h + a_2 2h \cdot h$$

$$\Rightarrow y_2 = y_0 + 2a_1 h + 2a_2 h^2$$

$$\Rightarrow y_2 = y_0 + 2\Delta y_0 + 2a_2 h^2$$

$$\Rightarrow 2a_2 h^2 = y_2 - y_0 - 2\Delta y_0$$

$$= y_2 - y_0 - 2(y_1 - y_0)$$

$$= y_2 - y_1 - (y_1 - y_0)$$

$$= \Delta y_1 - \Delta y_0$$

$$= \Delta^2 y_0$$

$$\therefore a_2 = \frac{\Delta^2 y_0}{2h^2} = \frac{\Delta^2 y_0}{2! h^2}$$

Similarly we get

$$a_3 = \frac{\Delta^3 y_0}{3! h^3}$$

.....

.....

$$a_n = \frac{\Delta^n y_0}{n! h^n}$$

Putting the value of $a_0, a_1, a_2, \dots, \dots, a_n$ in (1) we get

$$y_n(x) = y_0 + \frac{\Delta y_0}{h}(x - x_0) + \frac{\Delta^2 y_0}{2! h^2}(x - x_0)(x - x_1)$$

$$+ \frac{\Delta^3 y_0}{3! h^3}(x - x_0)(x - x_1)(x - x_2) + \dots + \frac{\Delta^n y_0}{n! h^n}(x - x_0)(x - x_1)(x - x_2) \dots (x - x_{n-1}) \dots (2)$$

Sitting $x = x_0 + ph$

Then $\frac{x-x_0}{h} = p$

$$\frac{x-x_1}{h} = \frac{x-(x_0+h)}{h} = \frac{x-x_0}{h} - \frac{h}{h} = p - 1$$

$$\frac{x-x_2}{h} = \frac{x-(x_0+2h)}{h} = \frac{x-x_0}{h} - \frac{2h}{h} = p - 2$$

Similarly,

$$\frac{x-x_{n-1}}{h} = \frac{x-(x_0+(n-1)h)}{h} = \frac{x-x_0}{h} - \frac{(n-1)h}{h} = p - (n-1)$$

Putting this value's in (2) we get

$$y_n(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots + \frac{p(p-1)(p-2)\dots(p-n+1)}{n!} \Delta^n y_0$$

Where $p = \frac{x-x_0}{h}$

Which is the Newton's forward difference interpolation formula and is useful for interpolation near the beginning of a set of tabulated value?

Ex 1. Find the cubic polynomial which takes the following values.

$y(0) = 1, y(1) = 0, y(2) = 1$ and $y(3) = 10$

Hence obtain $y(4)$

Solution. the difference table is as follows:

x	y	Δ	Δ^2	Δ^3
0	1			
1	0	-1		
2	1	1	2	
3	10	9	8	6

Here $h=1, x_0 = 0$

Hence $p = \frac{x-x_0}{h} = \frac{x-0}{1} = x$

Therefore from Newton's forward interpolation formula we get

$$\begin{aligned} y(x) &= 1 + x(-1) + \frac{x(x-1)}{2!} (2) + \frac{x(x-1)(x-2)}{3!} (6) \\ &= 1 - x + x^2 - x + x(x^2 - 3x + 2) \\ &= 1 - x + x^2 - x + x^3 - 3x^2 + 2x \\ &= x^3 - 2x^2 + 1 \end{aligned}$$

$$\begin{aligned} \text{Which is the required polynomial and } y(4) &= 4^3 - 2(4)^2 + 1 \\ &= 64 - 32 + 1 \\ &= 33 \end{aligned}$$

Ex 2. The population of a country in the decennial census were as under .Estimate The population for the year 1895.

Year : x	1891	1901	1911	1921	1931
Population : y (In thousands)	46	66	81	93	101

Solution. We know that the Newton's forward difference interpolation formula is ,

$$\begin{aligned} y_n(x) &= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots \\ &+ \frac{p(p-1)(p-2) + \dots + (p-n+1)}{n!} \Delta^n y_0 \dots \dots \dots (1) \end{aligned}$$

[Where , $p = \frac{x-x_0}{h}$]

Now, put

$$h=10, \quad x_0 = 1891, \quad x = 1895$$

$$p = \frac{1895-1891}{10} = 0.4$$

The difference table is:

x	y	Δ	Δ^2	Δ^3	Δ^4
1891	46				
1901	66	20			
1911	81	15	-5	2	
1921	93	12	-3	-1	-3
1931	101	8	-4		

Hence (1) gives,

$$\begin{aligned} y(1895) &= 46 + 0.4(20) + \frac{0.4(0.4-1)}{2!} (-5) + \frac{0.4(0.4-1)(0.4-2)}{3!} (2) + \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{4!} (-3) \\ &= 46 + 8 + \frac{0.4(-0.6)}{2} (-5) + \frac{0.4(-0.6)(-1.6)}{6} (2) + \frac{0.4(-0.6)(-1.6)(-2.6)}{24} (-3) \\ &= 54.85 \text{ thousands} \end{aligned}$$

Ex 3. Given, $\sin 45^\circ = 0.7071, \sin 50^\circ = 0.7660, \sin 55^\circ = 0.8192, \sin 60^\circ = 0.8660$
Find $\sin 52^\circ$ by using Newton's forward interpolation formula.

Solution. We know that the Newton's forward difference interpolation formula is,

$$\begin{aligned} y_n(x) &= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots + \frac{p(p-1)(p-2) + \dots + (p-n+1)}{n!} \Delta^n y_0 \dots \dots (1) \end{aligned}$$

[Where, $p = \frac{x-x_0}{h}$]

We have,

x	45°	50°	55°	60°
y	0.7071	0.7660	0.8192	0.8660

The difference table is as followed:

x	y	Δ	Δ^2	Δ^3
45°	0.7071			
50°	0.7660	0.0589		
55°	0.8192	0.0532	-0.0057	
60°	0.8660	0.0468	-0.0064	-0.0007

Here, $h = 5^\circ$, $x_0 = 45^\circ$, $x = 52^\circ$, $p = \frac{52^\circ - 45^\circ}{5^\circ} = 1.4$

So, from (1) we get,

$$\begin{aligned} y(52^\circ) &= 0.7071 + 1.4(0.0589) + \frac{1.4(1.4-1)}{2!}(-0.0057) \\ &\quad + \frac{1.4(1.4-1)(1.4-2)}{3!}(-0.0007) \\ &= 0.7071 + 0.08246 - 0.001567 + 0.0000392 \\ &= 0.7880032 \end{aligned}$$

Thus, $\sin 52^\circ = 0.788$ (approximately).

Newton's backward difference interpolation formula: Given the set of $(n+1)$ values viz. $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of x and y , it is required to find $y_n(x)$, a polynomial of the n th degree such that y and $y_n(x)$ agree at the tabulated values, let the values of x be equidistant. i.e.,

Let $x_i = x_0 + ih$ for $i = 1, 2, 3, \dots, n$

Since $y(x)$ is a polynomial of the n th degree, it may be written as

$$y_n(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + a_3(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \dots + a_n(x - x_n)(x - x_{n-1})(x - x_{n-2}) \dots (x - x_1) \dots (1)$$

Imposing now the condition that y and $y_n(x)$ should agree at the set of tabulated values.

When $x = x_n$ then $y_n = a_0$

When $x = x_{n-1}$ then $y_{n-1} = a_0 + a_1(x_{n-1} - x_n)$

$$\Rightarrow y_{n-1} = y_n + a_1(x_{n-1} - x_n)$$

$$\Rightarrow a_1(x_n - x_{n-1}) = y_n - y_{n-1}$$

$$\Rightarrow a_1 h = y_n - y_{n-1}$$

$$\Rightarrow a_1 = \frac{\nabla y_n}{h}$$

When $x = x_{n-2}$ then $y_{n-2} = a_0 + a_1(x_{n-2} - x_n) + a_2(x_{n-2} - x_n)(x_{n-2} - x_{n-1})$

$$\Rightarrow y_{n-2} = y_n - a_1(x_n - x_{n-2}) + a_2(x_n - x_{n-2})(x_{n-1} - x_{n-2})$$

$$\Rightarrow y_{n-2} = y_n - a_1 2h + a_2 2h \cdot h$$

$$\Rightarrow y_{n-2} = y_n - 2a_1 h + 2a_2 h^2$$

$$\Rightarrow 2a_2 h^2 = y_{n-2} + 2a_1 h - y_n$$

$$= y_{n-2} - 2(y_n - y_{n-1}) - y_n$$

$$= y_{n-2} + 2y_n - 2y_{n-1} - y_n$$

$$= y_{n-2} + y_n - 2y_{n-1}$$

$$= y_n - y_{n-1} - y_{n-1} + y_{n-2}$$

$$= (y_n - y_{n-1}) - (y_{n-1} - y_{n-2})$$

$$= \nabla y_n - \nabla y_{n-1}$$

$$= \nabla^2 y_n$$

$$\therefore a_2 = \frac{\nabla^2 y_n}{2!h^2}$$

Similarly we get,

$$a_3 = \frac{\nabla^3 y_n}{h^3 \cdot 3!}$$

.....

.....

$$a_n = \frac{\nabla^n y_n}{h^n \cdot n!}$$

Putting the value of $a_0, a_1, \dots, a_n$ in (1) we get

$$\begin{aligned} y_n(x) &= y_n + \frac{\nabla y_n}{h}(x - x_n) + \frac{\nabla^2 y_n}{h^2 \cdot 2!}(x - x_n)(x - x_{n-1}) \\ &\quad + \frac{\nabla^3 y_n}{h^3 \cdot 3!}(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \dots + \frac{\nabla^n y_n}{h^n \cdot n!}(x - x_n)(x - x_{n-1})(x - x_{n-2}) \dots (x - x_1) \dots (2) \end{aligned}$$

Sitting $x = x_n + ph$

$$\text{Then } \frac{x - x_n}{h} = p$$

$$\frac{x - x_{n-1}}{h} = \frac{x - (x_n - h)}{h} = \frac{x - x_n}{h} + \frac{h}{h} = p + 1$$

$$\frac{x - x_{n-2}}{h} = \frac{x - (x_n - 2h)}{h} = \frac{x - x_n}{h} + \frac{2h}{h} = p + 2$$

Similarly,

$$\frac{x - x_1}{h} = \frac{x - (x_n - (n-1)h)}{h} = \frac{x - x_n}{h} + \frac{(n-1)h}{h} = p + (n-1)$$

Putting this value's in (2) we get

$$y_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \cdots + \frac{p(p+1)(p+2)\cdots(p+n-1)}{n!} \nabla^n y_n$$

$$\text{Where } p = \frac{x - x_n}{h}$$

Which is the Newton's backward difference interpolation formula and is useful for interpolation near the end of a set of the value?

Ex4. The population of a country in the decennial census where as under .Estimate the population for the year 1925.

Year : x	1891	1901	1911	1921	1931
Population : y (In thousands)	46	66	81	93	101

Solution. Here the interpolation is desired at the end of the table and so we use Newton's backward differences interpolation formula.

We know that the Newton's backward difference interpolation formula is,

$$y_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \cdots + \frac{p(p+1)(p+2)\cdots(p+n-1)}{n!} \nabla^n y_n$$

Now, put

$$h = 10, \quad x_0 = 1931, \quad x = 1925$$

$$p = \frac{1931-1925}{10} = -0.6$$

The difference table is :

x	y	∇	∇^2	∇^3	∇^4
1891	46				
		20			
1901	66		-5		
		15		2	
1911	81		-3		-3
		12		-1	
1921	93		-4		
		8			
1931	101				

Hence (1) gives,

$$y(1925) = 101 + (-0.6)8 + \frac{-0.6(-0.6+1)}{2!}(-4) + \frac{-0.6(-0.6+1)(-0.6+2)}{3!}(-1) + \frac{-0.6(-0.6+1)(-0.6+2)(-0.6+3)}{4!}(-3) = 96.84 \text{ thousands}$$

Ex 5. The table below gives the value of $\tan(x)$ for $0.10 \leq x \leq 0.30$

X	0.10	0.15	0.20	0.25	0.30
---	------	------	------	------	------

Y=tan(x)	0.1003	0.1511	0.2027	0.2553	0.3093
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Find $\tan 0.26$

Solution. We know Newton's backward difference interpolation formula is,

$$y_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \cdots + \frac{p(p+1)(p+2)\cdots(p+n-1)}{n!} \nabla^n y_n \quad (1)$$

$$\text{Where } p = \frac{x - x_n}{h}$$

$$\text{Here } x=0.26, \quad x_n = 0.30, \quad h = 0.05$$

$$p = \frac{0.26 - 0.30}{0.05} = -0.8$$

The difference table is as follows,

x	Y	∇	∇^2	∇^3	∇^4
0.10	0.1003				
		0.0508			
0.15	0.1511		0.0008		
		0.0516		0.0002	
0.20	0.2027		0.0010		0.0002
		0.0526		0.0004	
0.25	0.25553		0.0014		
		0.0540			
0.30	0.3093				

Here(1) gives

$$y(0.26) = 0.3093 - 0.8(0.0540) + \frac{-0.8(-0.8+1)}{2!}(0.0014) + \frac{-0.8(-0.8+1)(-0.8+2)}{3!}(0.0004) + \frac{-0.8(-0.8+1)(-0.8+2)(-0.8+3)}{4!}(0.0002) = 0.2662 \quad (\text{Ans.})$$

Gauss's Forward Difference Interpolation Formula. We consider the following difference table in which the central ordinate is taken for convenience as y_0 corresponding to $x = x_0$.

x	y	Δ	Δ^2	Δ^3	Δ^4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$		$\Delta^4 y_{-4}$
		Δy_{-2}		$\Delta^3 y_{-3}$	
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_0	y_0	Δy_0	$\Delta^2 y_{-1}$	$\Delta^3 y_{-2}$	
			$\Delta^2 y_0$	$\Delta^3 y_{-1}$	
x_1	y_1	Δy_1		$\Delta^3 y_0$	$\Delta^4 y_{-1}$
			$\Delta^2 y_1$		$\Delta^4 y_0$
x_2	y_2		$\Delta^2 y_2$		$\Delta^4 y_1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

The differences used in this formula lie on the line shown in the table. The formula is, therefore, of the form.

$$y_p = y_0 + G_1 \Delta y_0 + G_2 \Delta^2 y_{-1} + G_3 \Delta^3 y_{-1} + G_4 \Delta^4 y_{-2} + \dots \dots \dots (i)$$

Where $G_1, G_2, G_3, \dots$ have to be determined. The y_p on the left side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$y_p = E^p y_0 = (1 + \Delta)^p y_0$$

$$= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

Similarly the right hand side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$\Delta^2 y_{-1} = \Delta^2 E^{-1} y_0 = \Delta^2 (1 + \Delta)^{-1} y_0 = \Delta^2 (1 - \Delta + \Delta^2 - \Delta^3 + \Delta^4 - \dots) y_0$$

$$\Rightarrow \Delta^2 y_{-1} = (\Delta^2 - \Delta^3 + \Delta^4 - \Delta^5 \dots) y_0$$

$$\Rightarrow \Delta^2 y_{-1} = \Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 \dots$$

$$\therefore \Delta^3 y_{-1} = \Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 \dots$$

Again

$$\Delta^4 y_{-2} = \Delta^4 E^{-2} y_0 = \Delta^4 (1 + \Delta)^{-2} y_0 = \Delta^4 (1 - 2\Delta + 3\Delta^2 - 4\Delta^3 + \dots) y_0$$

$$\Rightarrow \Delta^4 y_{-2} = \Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 \dots$$

Hence (i) give the identity

$$y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

$$= y_0 + G_1 \Delta y_0 + G_2 (\Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 \dots)$$

$$+ G_3 (\Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 \dots) + G_4 (\Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0) + \dots \dots (ii)$$

Equating the coefficients of $y_0, \Delta y_0, \Delta^2 y_0, \dots$ on both sides of (ii) we get

$$\begin{aligned} G_1 &= p \\ G_2 &= \frac{p(p-1)}{2!} \\ -G_2 + G_3 &= \frac{p(p-1)(p-2)}{3!} \\ \Rightarrow G_3 &= \frac{p(p-1)(p-2)}{3!} + \frac{p(p-1)}{2!} \\ \therefore G_3 &= \frac{(p+1)p(p-1)}{3!} \\ G_4 &= \frac{(p+1)p(p-1)(p-2)}{4!} \end{aligned}$$

Similarly

Hence (i) become

$$\begin{aligned} y_p &= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_{-1} \\ &\quad + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!} \Delta^4 y_{-2} + \dots \end{aligned}$$

Which is Gauss's forward formula.

Ex 6. Use Gauss's forward formula to find $f(x)$ when $x = 3.6$ from the data below

x	2.5	3.0	3.5	4.0	4.5
y	24.145	22.043	20.225	18.644	17.644

Soln. We know the gauss's forward formula is

$$y_p = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!} \Delta^4 y_{-2} + \dots \quad (i)$$

Here we take $x_0 = 3.5, x = 3.6, h = 0.5, p = \frac{x-x_0}{h} = \frac{3.6-3.5}{0.5} = 0.2$

The difference table is as follows:

x	y	Δ	Δ^2	Δ^3	Δ^4
2.5	24.145				
		-2.012			
3.0	22.043		.0284		
		-1.818		-0.47	
3.5	20.225	Δy_0	$\Delta^2 y_{-1}$	$\Delta^3 y_{-1}$	$\Delta^4 y_{-2}$
		-1.581	0.237	0.344	0.391
4.0	18.644		0.581 ₀		
		-1.000			
4.5	17.644				

Therefore $y_0 = 20.225$, $\Delta y_0 = -1.581$, $\Delta^2 y_{-1} = 0.237$, $\Delta^3 y_{-1} = 0.344$, $\Delta^4 y_{-2} = 0.391$

Hence from (i) we get

$$\begin{aligned}
 y_{0.2} &= y(3.6) \\
 &= 20.225 + 0.2(-1.581) + \frac{0.2(0.2-1)}{2!}(0.237) \\
 &\quad + \frac{(0.2+1)0.2(0.2-1)}{3!}(0.344) + \frac{(0.2+1)0.2(0.2-1)(0.2-2)}{4!}(0.391) \\
 &= 19.8844624 \text{ (Ans.)}
 \end{aligned}$$

Gauss's Backward Difference Interpolation Formula. We consider the following difference table in which the central ordinate is taken for convenience as y_0 corresponding to $x = x_0$.

x	y	Δ	Δ^2	Δ^3	Δ^4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$		$\Delta^4 y_{-4}$
		Δy_{-2}		$\Delta^3 y_{-3}$	
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_0	y_0	Δy_0	$\Delta^2 y_{-1}$	$\Delta^3 y_{-1}$	$\Delta^4 y_{-2}$
		Δy_1	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_{-1}$
x_1	y_1		$\Delta^2 y_1$		$\Delta^4 y_0$
x_2	y_2		$\Delta^2 y_2$		$\Delta^4 y_1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

The differences used in this formula lie on the line shown in the table. The formula is, therefore, of the form.

$$y_p = y_0 + G_1 \Delta y_{-1} + G_2 \Delta^2 y_{-1} + G_3 \Delta^3 y_{-2} + G_4 \Delta^4 y_{-2} + \dots \dots \dots (i)$$

Where $G_1, G_2, G_3, \dots$ have to be determined. The y_p on the left side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$\begin{aligned}
 y_p &= E^p y_0 = (1 + \Delta)^p y_0 \\
 &= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 \\
 &\quad + \dots
 \end{aligned}$$

Similarly the right hand side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$\Delta y_{-1} = \Delta E^{-1} y_0 = \Delta(1 + \Delta)^{-1} y_0 = \Delta(1 - \Delta + \Delta^2 - \Delta^3 + \Delta^4 - \dots) y_0$$

$$\Delta y_{-1} = (\Delta - \Delta^2 + \Delta^3 - \Delta^4 \dots) y_0$$

$$\Rightarrow \Delta y_{-1} = \Delta y_0 - \Delta^2 y_0 + \Delta^3 y_0 - \Delta^4 y_0 \dots$$

$$\therefore \Delta^2 y_{-1} = \Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 \dots$$

Again

$$\Delta^3 y_{-2} = \Delta^3 E^{-2} y_0 = \Delta^3 (1 + \Delta)^{-2} y_0 = \Delta^3 (1 - 2\Delta + 3\Delta^2 - 4\Delta^3 + \dots) y_0$$

$$\Rightarrow \Delta^3 y_{-2} = \Delta^3 y_0 - 2\Delta^4 y_0 + 3\Delta^5 y_0 - 4\Delta^6 y_0 \dots$$

$$\therefore \Delta^4 y_{-2} = \Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 \dots$$

Hence (i) give the identity

$$\begin{aligned}
 y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots \\
 = y_0 + G_1 (\Delta y_0 - \Delta^2 y_0 + \Delta^3 y_0 - \Delta^4 y_0 \dots) + G_2 (\Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 \dots) \\
 + G_3 (\Delta^3 y_0 - 2\Delta^4 y_0 + 3\Delta^5 y_0 - 4\Delta^6 y_0 \dots) + G_4 (\Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 \dots) + \dots
 \end{aligned} \quad (ii)$$

Equating the coefficients of $y_0, \Delta y_0, \Delta^2 y_0, \dots$ on both sides of (ii) we get

$$\begin{aligned}
 G_1 &= p \\
 -G_1 + G_2 &= \frac{p(p-1)}{2!} \\
 \Rightarrow G_2 &= \frac{p(p-1)}{2!} + p \\
 \therefore G_2 &= \frac{p(p+1)}{2!}
 \end{aligned}$$

Similarly

$$\begin{aligned}
 G_3 &= \frac{(p+1)p(p-1)}{3!} \\
 G_4 &= \frac{(p+2)(p+1)p(p-1)}{4!}
 \end{aligned}$$

Hence (i) becomes

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_{-1} + \frac{p(p+1)}{2!} \Delta^2 y_{-1} \\
 &\quad + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-2} + \frac{(p+2)(p+1)p(p-1)}{4!} \Delta^4 y_{-2} + \dots
 \end{aligned}$$

Which is Gauss's backward formula.

Ex 7. Interpolate by means of Gauss's backward formula the population for the year 1936, Given the following table

Year	1901	1911	1921	1931	1941	1951
Population (in thousand)	12	15	20	27	39	52

Soln. We know the gauss's backward formula is

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_{-1} + \frac{p(p+1)}{2!} \Delta^2 y_{-1} \\
 &\quad + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-2} + \frac{(p+2)(p+1)p(p-1)}{4!} \Delta^4 y_{-2} + \dots
 \end{aligned} \quad \dots \dots \dots (i)$$

Here we take $x_0 = 1931$, $x = 1936$, $h = 10$, $p = \frac{x-x_0}{h} = \frac{1936-1931}{10} = 0.5$

The difference table is as follows:

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5
1901	12					
		3				
1911	15		2			
		5		0		
1921	20		2		3	
		7		3		
1931	27		5		-7	
		12		-4		-10
1941	39		1			
		13				
1951	52					

Therefore $y_0 = 27$, $\Delta y_{-1} = 7$, $\Delta^2 y_{-1} = 5$, $\Delta^3 y_{-2} = 3$, $\Delta^4 y_{-2} = -7$, $\Delta^5 y_{-3} = -10$

Hence from (i) we get

$$y_{0.5} = y(1936)$$

$$\begin{aligned}
 &= 27 + 0.5(7) + \frac{0.5(0.5+1)}{2!}(5) + \frac{(0.5+1)0.5(0.5-1)}{3!}(3) \\
 &+ \frac{(0.5+2)(0.5+1)0.5(0.5-1)}{4!}(-7) \\
 &+ \frac{(0.5+2)(0.5+1)0.5(0.5-1)(0.5-2)}{5!}(-10) \\
 &= 27 + 3.5 + 1.875 - 0.1875 + 0.2734 - 0.1172 \\
 &= 32.3437 \text{ (Thousand) (Ans.)}
 \end{aligned}$$

Stirling's interpolation Formula. We know Gauss's forward formula is

$$\begin{aligned}
 y_p &= y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} \\
 &+ \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \dots
 \end{aligned}
 \tag{i}$$

And Gauss's backward formula is

$$\begin{aligned}
 y_p &= y_0 + p\Delta y_{-1} + \frac{p(p+1)}{2!}\Delta^2 y_{-1} \\
 &+ \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-2} + \frac{(p+2)(p+1)p(p-1)}{4!}\Delta^4 y_{-2} + \dots
 \end{aligned}
 \tag{ii}$$

Now taking the mean of (i) and (ii) we obtain

$$\begin{aligned}
 y_p &= y_0 + p\frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} + \frac{P^2(p^2-1)}{4!}\Delta^4 y_{-2} \\
 &+ \dots
 \end{aligned}$$

Which is called the Stirling's formula.

Ex 8. Use Stirling's formula to find $f(35)$ where $f(20) = 512$, $f(30) = 439$, $f(40) = 346$, $f(50) = 243$.

Soln. We know the Stirling's formula is

$$\begin{aligned}
 y_p &= y_0 + p\frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} + \frac{P^2(p^2-1)}{4!}\Delta^4 y_{-2} \\
 &+ \dots
 \end{aligned}
 \tag{i}$$

The difference table is as follows:

x	$f(x)$	Δ	Δ^2	Δ^3
20	512			
		-73		
30	439		-20	
		-93		10
40	346		-10	
		-103		
50	243			

Here we take $x_0 = 30$, $x = 35$, $h = 10$, $p = \frac{x-x_0}{h} = \frac{35-30}{10} = \frac{5}{10} = 0.5$

$$\begin{aligned}
 \text{Therefore, } y_{0.5} &= y(35) = 439 + 0.5\frac{-73-93}{2} + \frac{0.5^2}{2!}(-20) + \frac{0.5(0.5^2-1)}{3!}\frac{10+0}{2} \\
 &= 395 \text{ (Ans.)}
 \end{aligned}$$

Bessel's interpolation formula. This formula uses the differences as shown in the following table where brackets mean that the average of the values have to be taken.

x	y	Δ	Δ^2	Δ^3	Δ^4
x_{-2}	y_{-2}		$\Delta^2 y_{-3}$		$\Delta^4 y_{-4}$
		Δy_{-2}		$\Delta^3 y_{-3}$	
x_{-1}	y_{-1}		$\Delta^2 y_{-2}$		$\Delta^4 y_{-3}$
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_0	$\left(\begin{array}{c} y_0 \\ y_1 \end{array} \right)$	Δy_0	$\left(\begin{array}{c} \Delta^2 y_{-1} \\ \Delta^2 y_0 \end{array} \right)$	$\Delta^3 y_{-1}$	$\left(\begin{array}{c} \Delta^3 y_{-2} \\ \Delta^3 y_{-1} \end{array} \right)$
x_1		Δy_1		$\Delta^3 y_0$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$

Here Bessel's formula can be assumed in the form,

$$y_p = \frac{y_0 + y_1}{2} + B_1 \Delta y_0 + B_2 \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + B_3 \Delta^3 y_0 + B_4 \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} + \dots$$

$$\Rightarrow y_p = y_0 + (B_1 + \frac{1}{2}) \Delta y_0 + B_2 \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + B_3 \Delta^3 y_0 + B_4 \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} + \dots$$

.....(i)

Where $B_1, B_2, B_3, \dots$ have to be determined. The y_p on the left side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$y_p = E^p y_0 = (1 + \Delta)^p y_0$$

$$= y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

Similarly the right hand side can be expressed in terms of $y_0, \Delta y_0$ and higher order differences of y_0 as follows.

$$\Delta^2 y_{-1} = \Delta^2 E^{-1} y_0 = \Delta^2 (1 + \Delta)^{-1} y_0 = \Delta^2 (1 - \Delta + \Delta^2 - \Delta^3 + \Delta^4 - \dots) y_0$$

$$\Delta^2 y_{-1} = (\Delta^2 - \Delta^3 + \Delta^4 - \Delta^5 + \dots) y_0$$

$$\Rightarrow \Delta^2 y_{-1} = \Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 + \dots$$

$$\Rightarrow \Delta^2 y_{-1} = \Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 + \dots$$

$$\therefore \Delta^3 y_{-1} = \Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 + \dots$$

$$\therefore \Delta^4 y_{-1} = \Delta^4 y_0 - \Delta^5 y_0 + \Delta^6 y_0 - \Delta^7 y_0 + \dots$$

Again

$$\Delta^4 y_{-2} = \Delta^4 E^{-2} y_0 = \Delta^4 (1 + \Delta)^{-2} y_0 = \Delta^4 (1 - 2\Delta + 3\Delta^2 - 4\Delta^3 + \dots) y_0$$

$$\Rightarrow \Delta^4 y_{-2} = \Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 + \dots$$

Hence (i) give the identity

$$y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

$$= y_0 + \left(B_1 + \frac{1}{2}\right) \Delta y_0 + B_2 \frac{((\Delta^2 y_0 - \Delta^3 y_0 + \Delta^4 y_0 - \Delta^5 y_0 + \dots) + \Delta^2 y_0)}{2}$$

$$+ B_3 (\Delta^3 y_0 - \Delta^4 y_0 + \Delta^5 y_0 - \Delta^6 y_0 + \dots)$$

$$+ B_4 \frac{(\Delta^4 y_0 - 2\Delta^5 y_0 + 3\Delta^6 y_0 - 4\Delta^7 y_0 + \dots) + (\Delta^4 y_0 - \Delta^5 y_0 + \Delta^6 y_0 - \Delta^7 y_0 + \dots)}{2} + \dots$$

.....(ii)

Equating the coefficient of $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0, \dots$ on both sides of (ii), we obtain

$$p = B_1 + \frac{1}{2} \Rightarrow B_1 = p - \frac{1}{2}$$

Again

$$B_2 = \frac{p(p-1)}{2!}$$

And

$$-\frac{B_2}{2} + B_3 = \frac{p(p-1)(p-2)}{3!}$$

$$\Rightarrow B_3 = \frac{p(p-1)(p-2)}{3!} + \frac{p(p-1)}{4}$$

$$\Rightarrow B_3 = \frac{p(p-1)(p-\frac{1}{2})}{3!}$$

Similarly

$$B_4 = \frac{(p+1)p(p-1)(p-2)}{4!}$$

Hence from (i)

$$y_p = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \cdot \frac{(\Delta^2 y_{-1} + \Delta^2 y_0)}{2} + \frac{p(p-1)(p-\frac{1}{2})}{3!} \cdot \Delta^3 y_{-1}$$

$$+ \frac{(p+1)p(p-1)(p-2)}{4!} \cdot \frac{(\Delta^4 y_{-2} + \Delta^4 y_{-1})}{2} + \dots$$

Which is the Bessel's formula,

Ex 9. The following table gives readings of the temperature θ of a metal recorded in given time (t).

Time t	2	3	4	5	6
Temperature θ	61.87	54.08	47.03	40.65	34.88

Using Bessel's formula find temperature θ at time $t = 4.3$.

Ans. $\approx 45.04838 \approx 45.05$

Interpolation with unequal intervals

Divided Differences. Let $f(x_0), f(x_1), \dots, f(x_n)$ be the entries corresponding to the arguments $x_0, x_1, \dots, x_n$ where the intervals $x_1 - x_0, x_2 - x_1, \dots, x_n - x_{n-1}$ may not be equal. Then the first divided difference of $f(x)$ for the arguments x_0, x_1 is defined as $\frac{f(x_1) - f(x_0)}{x_1 - x_0}$ and is denoted by $f(x_0, x_1)$, that is, $f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$

Similarly the other first divided differences of $f(x)$ for the arguments $x_1, x_2; x_2, x_3; \dots, x_{n-1}, x_n$ are

$$f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{x_3 - x_2}$$

.....

$$f(x_{n-1}, x_n) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

The second divided difference of $f(x)$ for the three arguments x_0, x_1 and x_2 is defined as

$$f(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$$

The n th divided difference is given by $f(x_0, x_1, x_2, \dots, x_n) = \frac{f(x_1, x_2, \dots, x_n) - f(x_0, x_1, \dots, x_{n-1})}{x_n - x_0}$

Note: If two of the arguments coincide the divided difference can be given a meaning assigned by taking the limit. Thus

$$\begin{aligned} f(x_0, x_0) &= \lim_{\epsilon \rightarrow 0} (x_0, x_0 + \epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \frac{f(x_0 + \epsilon) - f(x_0)}{(x_0 + \epsilon) - x_0} \\ &= \lim_{\epsilon \rightarrow 0} \frac{f(x_0 + \epsilon) - f(x_0)}{\epsilon} \\ &= f'(x_0) \end{aligned}$$

Similarly $f(x_0, x_0, \dots, x_0) = \frac{1}{r!} f^{(r)}(x_0)$
($r+1$) arguments

Theorem: The divided difference are symmetrical in all its arguments, That is independent of the order of the arguments.

Proof: We have

$$\begin{aligned} f(x_0, x_1) &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_0) - f(x_1)}{x_0 - x_1} = f(x_1, x_0) \\ &= \frac{f(x_0)}{x_0 - x_1} + \frac{f(x_1)}{x_1 - x_0} \\ &= \sum \frac{f(x_i)}{x_i - x_j}, \text{ Showing that } f(x_0, x_1) \text{ is symmetrical in } x_0, x_1, \end{aligned}$$

$$\begin{aligned} \text{Again } f(x_0, x_1, x_2) &= \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0} \\ &= \frac{1}{x_2 - x_0} \left[\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] \\ &= \frac{1}{x_2 - x_0} \left[\frac{f(x_2)}{x_2 - x_1} - \frac{f(x_1)}{x_1 - x_0} - \frac{f(x_1)}{x_1 - x_0} + \frac{f(x_0)}{x_1 - x_0} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{x_2 - x_0} \left[\frac{f(x_2)}{x_2 - x_1} - f(x_1) \left(\frac{1}{x_2 - x_1} - \frac{1}{x_1 - x_0} \right) + \frac{f(x_0)}{x_1 - x_0} \right] \\
&= \frac{1}{x_2 - x_0} \left[\frac{f(x_2)}{x_2 - x_1} - f(x_1) \frac{x_1 - x_0 + x_2 - x_1}{(x_2 - x_1)(x_1 - x_0)} + \frac{f(x_0)}{x_1 - x_0} \right] \\
&= \frac{1}{x_2 - x_0} \left[\frac{f(x_2)}{x_2 - x_1} - \frac{f(x_1)(x_2 - x_0)}{(x_2 - x_1)(x_1 - x_0)} + \frac{f(x_0)}{x_1 - x_0} \right] \\
&= \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} + \frac{f(x_1)}{(x_1 - x_2)(x_1 - x_0)} + \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} \\
&= \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} \\
&= \sum \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)}, \text{ Showing that } f(x_0, x_1, x_2) \text{ is symmetrical in } x_0, x_1, x_2.
\end{aligned}$$

Let us assume similar symmetrical expressions for the (n-1)th divided differences. That is, let us assume that

$$\begin{aligned}
f(x_0, x_1, \dots, x_{n-1}) &= \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_{n-1})} \\
&\quad + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_{n-1})} + \dots \\
&\quad + \frac{f(x_{n-1})}{(x_{n-1} - x_0)(x_{n-1} - x_1) \dots (x_{n-1} - x_{n-2})} \\
&= \sum \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_{n-1})}
\end{aligned}$$

And similar expressions for the other (n-1)th divided difference.

$$\begin{aligned}
\text{Then } f(x_0, x_1, \dots, x_n) &= \frac{f(x_0, x_1, \dots, x_{n-1}) - f(x_1, x_2, \dots, x_n)}{x_0 - x_n} \\
&= \frac{1}{x_0 - x_n} \left[\left\{ \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_{n-1})} + \frac{f(x_1)}{(x_1 - x_0) \dots (x_1 - x_{n-1})} + \dots \frac{f(x_{n-1})}{(x_{n-1} - x_0) \dots (x_{n-1} - x_{n-2})} \right\} - \right. \\
&\quad \left. \left\{ \frac{f(x_1)}{(x_1 - x_2) \dots (x_1 - x_n)} + \frac{f(x_2)}{(x_2 - x_1) \dots (x_2 - x_n)} + \dots + \frac{f(x_n)}{(x_n - x_1) \dots (x_n - x_{n-1})} \right\} \right] \\
&= \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_n)} + \frac{f(x_1)}{(x_1 - x_0) \dots (x_1 - x_n)} + \dots \frac{f(x_n)}{(x_n - x_0) \dots (x_n - x_{n-1})} \\
&= \sum \frac{f(x_0)}{(x_0 - x_1) \dots (x_0 - x_n)}
\end{aligned}$$

showing that the nth divided difference $f(x_0, x_1, \dots, x_n)$ is also symmetrical in $x_0, x_1, \dots, x_n$ and thus completing the proof of the theorem by mathematical induction.

Theorem: The nth divided differences of a polynomial of the nth degree are constant.

Proof: First consider the function $f(x) = x^n$. The first divided differences of this function are given by

$$\begin{aligned}
f(x_r, x_{r+1}) &= \frac{f(x_{r+1}) - f(x_r)}{x_{r+1} - x_r} \\
&= \frac{x_{r+1}^n - x_r^n}{x_{r+1} - x_r} \\
&= x_{r+1}^{n-1} + x_r x_{r+1}^{n-2} + \dots + x_{r+1} x_r^{n-2} + x_r^{n-1}
\end{aligned}$$

Which is a homogeneous expression of degree n-1 in x_{r+1} and x_r . The second divided differences are given by

$$\begin{aligned}
f(x_r, x_{r+1}, x_{r+2}) &= \frac{f(x_r, x_{r+1}) - f(x_{r+1}, x_{r+2})}{x_r - x_{r+2}} \\
&= \frac{\frac{f(x_{r+1}, x_{r+2}) - f(x_r, x_{r+1})}{x_{r+2} - x_r}}{x_r - x_{r+2}} \\
&= \frac{1}{(x_{r+2} - x_r)} [f(x_{r+1}, x_{r+2}) - f(x_r, x_{r+1})] \\
&= \frac{1}{x_{r+2} - x_r} \left[\frac{f(x_{r+2}) - f(x_{r+1})}{x_{r+2} - x_{r+1}} - \frac{f(x_{r+1}) - f(x_r)}{x_{r+1} - x_r} \right] \\
&= \frac{1}{x_{r+2} - x_r} \left[\frac{x_{r+2}^n - x_{r+1}^n}{x_{r+2} - x_{r+1}} - \frac{x_{r+1}^n - x_r^n}{x_{r+1} - x_r} \right] \\
&= \frac{1}{x_{r+2} - x_r} [(x_{r+2}^{n-1} + x_{r+1} x_{r+2}^{n-2} + \dots + x_{r+1}^{n-2} x_{r+2} + x_{r+2}^{n-1}) - (x_{r+1}^{n-1} + x_r x_{r+1}^{n-2} + \dots \\
&\quad \dots + x_r^{n-2} x_{r+1} + x_r^{n-1})] \\
&= \frac{x_{r+1}^{n-1} - x_r^{n-1}}{x_{r+2} - x_r} + x_{r+1} \frac{x_{r+2}^{n-2} - x_r^{n-2}}{x_{r+2} - x_r} + x_{r+1}^{n-2} \frac{x_{r+2} - x_r}{x_{r+2} - x_r} \\
&= (x_{r+2}^{n-2} + x_r x_{r+2}^{n-3} + \dots + x_r^{n-2}) + x_{r+1} (x_{r+2}^{n-3} + \dots + x_r^{n-3}) + \dots + x_{r+1}^{n-2}
\end{aligned}$$

Which is a homogeneous expression of degree n-2 in x_r, x_{r+1}, x_{r+2} by induction it can be shown that $f(x_r, x_{r+1}, \dots, x_{r+m})$ is a homogeneous expression of degree n-m.

In particular, The nth divided difference of $f(x) = x^n$ is an expression of degree zero, it is a constant, and is therefore independent of the values of $x_r, x_{r+1}, \dots, x_{r+n}$

Since the nth divided difference of x^n are constant, Therefore the divided differences of x^n of order greater than n will all be zero.

If $f(x) = ax^n$, where a is a constant, Then the nth divided difference of $f(x) = a$

(The nth divided difference of x^n) Which is a constant.

Hence if $f(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$ be a polynomial of degree n, Then the nth divided differences of all the terms except $a_0 x^n$ will be zero. and so the nth divided difference of the whole polynomial will be constant.

Newton's interpolation formula for unequal intervals.

Or. Newton's interpolation formula for divided difference.

Let $f(x_0), f(x_1), \dots, f(x_n)$ be the values of $f(x)$ corresponding to the arguments $x_0, x_1, \dots, x_n$, not necessarily equally spaced. From the definition of divided differences,

$$f(x, x_0) = \frac{f(x) - f(x_0)}{x - x_0}$$

$$\text{Or. } f(x) = f(x_0) + (x - x_0)f(x, x_0) \quad (1)$$

$$\text{Also } f(x, x_0, x_1) = \frac{f(x, x_0) - f(x_0, x_1)}{x - x_1}$$

$$\Rightarrow f(x, x_0) = f(x_0, x_1) + \dots (x - x_1)f(x, x_0, x_1) \dots \dots \dots (2)$$

Similarly

$$f(x, x_0, x_1) = f(x_0, x_1, x_2) + (x - x_2)f(x, x_0, x_1, x_2) \dots \dots \dots (3)$$

.....

$$f(x, x_0, x_1, \dots \dots x_{n-1}) = f(x_0, x_1, \dots \dots x_n) + (x - x_n)f(x, x_0, x_1, \dots \dots x_n) \dots \dots \dots (4)$$

Multiplying the equation (2) by $(x - x_0)$ (3) by $(x - x_0)(x - x_1)$ and so on and finally the equation (4) by $(x - x_0)(x - x_1) \dots \dots (x - x_{n-1})$ and adding to the equation (1), we have

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + \dots \dots \dots + (x - x_0)(x - x_1) \dots \dots (x - x_{n-1})f(x_0, x_1, \dots \dots x_n) + R_n$$

Where R_n is given by

$$R_n = (x - x_0)(x - x_1) \dots \dots \dots (x - x_n)f(x, x_0, x_1, \dots \dots \dots x_n) \quad \text{Assuming that } f(x) \text{ is a polynomial of degree } n, \text{ then } f(x, x_0, \dots \dots x_n)$$

Vanishes, so that

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + \dots \dots \dots + (x - x_0)(x - x_1) \dots \dots (x - x_{n-1})f(x_0, x_1, \dots \dots x_n)$$

This formula is called Newton's divided difference interpolation formula.

Ex 1. Fit a polynomial which passes through the points (1,4), (3,7), (4,8) and (6,11) by using Newton's divided difference formula. Also find the value of y at x=2.

Soln. The difference table is as follows.

x	f(x)	1 st diff	2 nd diff	3 rd diff
1	4			
3	7	$\frac{7-4}{3-1} = 1.5$	$\frac{1-1.5}{4-1} = -.167$	
4	8	$\frac{8-7}{4-3} = 1$		$\frac{.167 + .167}{6-1} = .067$
6	11	$\frac{11-8}{6-4} = 1.5$	$\frac{1.5-1}{6-3} = .167$	

Here $f(1) = 4, f(1,3) = 1.5, f(1,3,4) = -.167, f(1,3,4,6) = .067$

Therefore from Newton's divided difference formula we get

$$f(x) = f(1) + (x - 1)f(1,3) + (x - 1)(x - 3)f(1,3,4) + (x - 1)(x - 3)(x - 4)f(1,3,4,6) \\ = 4 + 1.5(x - 1) - .167(x - 1)(x - 3) + .067(x - 1)(x - 3)(x - 4)$$

Which is the required polynomial

Now put $x = 2$, we get

$$y = f(2) \\ = 4 + 1.5(2 - 1) - .167(2 - 1)(2 - 3) + .067(2 - 1)(2 - 3)(2 - 4) \\ = 5.799 \quad \text{Ans.}$$

Ex 2. If $(x) = \frac{1}{x^2}$, find $f(a, b), f(a, b, c)$ and $f(a, b, c, d)$

Soln : We have

$$f(a, b) = \frac{f(b) - f(a)}{b - a} = \frac{\frac{1}{b^2} - \frac{1}{a^2}}{b - a} = \frac{\frac{a^2 - b^2}{a^2 b^2}}{b - a} = \frac{(a + b)(a - b)}{a^2 b^2 (b - a)} = -\frac{a + b}{a^2 b^2} \dots \dots \dots (1)$$

$$\text{Again } f(a, b, c) = \frac{f(b, c) - f(a, b)}{c - a} \\ = \frac{-\frac{b + c}{b^2 c^2} + \frac{a + b}{a^2 b^2}}{c - a} \\ = \frac{1}{c - a} \cdot \frac{c^2(a + b) - a^2(b + c)}{a^2 b^2 c^2} \\ = \frac{1}{c - a} \cdot \frac{(c^2 - a^2)b + c^2 a - a^2 c}{a^2 b^2 c^2} \\ = \frac{1}{c - a} \cdot \frac{(c - a)\{(c + a)b + ca\}}{a^2 b^2 c^2} = \frac{ab + bc + ca}{a^2 b^2 c^2}$$

$$\text{Similarly } f(a, b, c, d) = -\frac{abc + bcd + acd + abd}{a^2 b^2 c^2 d^2}$$

Ex 3. Show that $\frac{\Delta^2}{yz}x^3 = x + y + z$

Soln: $f(x) = x^3$

Now $\frac{\Delta}{y}f(x) = f(x, y) = \frac{f(y) - f(x)}{y - x}$

$$= \frac{y^3 - x^3}{y - x}$$

$$= y^2 + xy + x^2$$

Similarly $\frac{\Delta}{y}f(z) = z^2 + yz + y^2$

Now $\frac{\Delta^2}{yz}f(x) = \frac{\Delta^2}{yz}x^3$

$$= f(x, y, z)$$

$$= \frac{f(y, z) - f(x, y)}{z - x}$$

$$= \frac{y^2 + yz + z^2 - x^2 - xy - y^2}{z - x}$$

$$= \frac{z^2 - x^2 + yz - xy}{z - x}$$

$$= \frac{(z - x)(z + x + y)}{z - x}$$

$$= x + y + z \quad \text{(Proved).}$$

Lagranges Interpolation formula for unequal intervals:

Let $f(x)$ denote a polynomial of the n th degree which takes the values $y_0, y_1, \dots, y_n$ when x has the values $x_0, x_1, x_2, \dots, x_n$ respectively. Then the $(n+1)$ th differences of this polynomial are zero. Hence

$$f(x, x_0, x_1, \dots, x_n) = 0 \quad \dots \dots \dots (1)$$

But we have

$$f(x, x_0, x_1, \dots, x_n)$$

$$= \frac{y}{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)}$$

$$+ \frac{y_0}{(x_0 - x)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)}$$

$$+ \frac{y_1}{(x_1 - x)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots \dots \dots$$

$$+ \frac{y_n}{(x_n - x)(x_n - x_0) \dots (x_n - x_{n-1})}$$

$\therefore$ using (1), we get

$$\begin{aligned} \Rightarrow & \frac{y}{(x - x_0)(x - x_1) \dots (x - x_n)} \\ & + \frac{y_0}{(x_0 - x)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} \\ & + \frac{y_1}{(x_1 - x)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots \dots \dots \\ & + \frac{y_n}{(x_n - x)(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} = 0 \end{aligned}$$

$$\Rightarrow \frac{y}{(x - x_0)(x - x_1) \dots (x - x_n)} = \frac{y_0}{(x - x_0)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} + \frac{y_1}{(x - x_1)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} + \dots \dots \dots + \frac{y_n}{(x - x_n)(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})}$$

$$\Rightarrow y = \frac{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)}{(x - x_0)(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 +$$

$$\frac{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)}{(x - x_1)(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 + \dots \dots \dots$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_2) \dots (x - x_n)}{(x - x_n)(x_n - x_0)(x_n - x_2) \dots (x_n - x_{n-1})} y_n$$

$$\therefore y = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1 + \dots \dots \dots$$

$$+ \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n$$

Which is Lagrange's formula.

Ex 4. Apply Lagrange's Formula Calculate $f(1.5)$ by using the following table.

	x_0	x_1	x_2	x_3	x_4	x_5
x	1.00	1.20	1.40	1.60	1.80	2.00
$f(x)$	.2420	.1942	.1497	.1109	.0790	.0540

Soln. Using Lagrange formula we get,

$$\begin{aligned} f(1.5) &= \frac{(1.5 - 1.2)(1.5 - 1.4)(1.5 - 1.6)(1.5 - 1.8)(1.5 - 2)}{(1 - 1.2)(1 - 1.4)(1 - 1.6)(1 - 1.8)(1 - 2)} (.2420) \\ &+ \frac{(1.5 - 1)(1.5 - 1.4)(1.5 - 1.6)(1.5 - 1.8)(1.5 - 2)}{(1.2 - 1)(1.2 - 1.4)(1.2 - 1.6)(1.2 - 1.8)(1.2 - 2)} (.1942) \\ &+ \frac{(1.5 - 1)(1.5 - 2)(1.5 - 1.6)(1.5 - 1.8)(1.5 - 2)}{(1.4 - 1)(1.4 - 1.2)(1.4 - 1.6)(1.4 - 1.8)(1.4 - 2)} (.1497) \\ &+ \frac{(1.5 - 1)(1.5 - 1.2)(1.5 - 1.4)(1.5 - 1.8)(1.5 - 2)}{(1.6 - 1)(1.6 - 1.2)(1.6 - 1.4)(1.6 - 1.8)(1.6 - 2)} (.1109) \\ &+ \frac{(1.5 - 1)(1.5 - 1.2)(1.5 - 1.4)(1.5 - 1.6)(1.5 - 2)}{(1.8 - 1)(1.8 - 1.2)(1.8 - 1.4)(1.8 - 1.6)(1.8 - 2)} (.0790) \\ &+ \frac{(1.5 - 1)(1.5 - 1.2)(1.5 - 1.4)(1.5 - 1.6)(1.5 - 1.8)}{(2 - 1)(2 - 1.2)(2 - 1.4)(2 - 1.6)(2 - 1.8)} (.0540) \\ &= .002835937 - .018964843 + .087714843 \\ &\quad - .064980468 - .007714843 + .000632812 \\ &= .129484375 \quad \text{Ans.} \end{aligned}$$

Calculus of finite differences

The calculus of finite differences is the study of changes that take place in the value of the function or the dependent variable, say $y = f(x)$, with respect to the finite changes in the independent variable x .

Let $y = f(x) = x^2$ be the function of the independent variable x . If there is a finite increase in x , say Δx , then $f(x) = x^2$ will be increased to $f(x + \Delta x) = (x + \Delta x)^2$ bearing the increment $f(x + \Delta x) - f(x) = (x + \Delta x)^2 - x^2$.

Such finite changes are studied in the calculus of finite differences.

- **The Operators E , Δ and ∇ .**

The operator E : Let $y = f(x)$ be any function of x . By operating E on $f(x)$ we mean to simply give an increment to the value of x in the function $f(x)$. If this increment be denoted by h , then the operation of E on $f(x)$ means that put $x + h$ in the function $f(x)$ wherever there is x i.e.

$$Ef(x) = f(x + h).$$

Here we should note that $Ef(x)$ does not imply the multiplication of E and $f(x)$ but it implies that E is operated on $f(x)$. The operator E is known as the shift operator. By $E^2 f(x)$ we mean that the operator E^2 is applied twice on the function $f(x)$ i.e.

$$\begin{aligned} E^2 f(x) &= E\{Ef(x)\} \\ &= E\{f(x + h)\} \\ &= f(x + h + h) \\ &= f(x + 2h). \end{aligned}$$

Similarly $E^n f(x)$ means that the operator E is applied n times on the function $f(x)$. i.e.

$$E^n f(x) = f(x + nh).$$

Remark. The operator E^{-1} is the inverse operator of the operator E and is defined as

$$\begin{aligned} E^{-1} f(x) &= f(x + (-1)h) \\ &= f(x - h) \end{aligned}$$

The operator Δ : Let $y = f(x)$ be any function of x . Let the consecutive values of x be a , $a + h$, $a + 2h$, , $a + nh$ differing by h . Then the corresponding values of y are $f(a)$, $f(a + h)$, $f(a + 2h)$, , $f(a + nh)$.

The independent variable x known as argument and the dependent variable y is known as entry. Thus we are given a set values of argument and entry.

The difference $f(a + h) - f(a)$ is called the first forward difference of the function $f(x)$ at the point $x = a$ and we denote by $\Delta f(a)$, i.e.,

$$\Delta f(a) = f(a + h) - f(a)$$

Similarly,

$$\Delta f(a + h) = f(a + 2h) - f(a + h)$$

The general formula is

$$\Delta f(a + nh) = f(a + (n + 1)h) - f(a + nh)$$

The operator Δ is called the forward difference operator and the differences $\Delta f(a)$, $\Delta f(a+h)$, $\Delta f(a+2h)$, etc. are called first forward differences.

The difference $\Delta f(a+h) - \Delta f(a)$ is known as the second forward difference of $f(x)$ at the point $x=a$ and is denoted by $\Delta^2 f(a)$, i.e.,

$$\begin{aligned}\Delta^2 f(a) &= \Delta f(a+h) - \Delta f(a) \\ &= \{f(a+2h) - f(a+h)\} - \{f(a+h) - f(a)\} \\ &= f(a+2h) - 2f(a+h) + f(a)\end{aligned}$$

Similarly $\Delta^3 f(a) = \Delta^2 f(a+h) - \Delta^2 f(a)$.

In general $\Delta^n f(a) = \Delta^{n-1} f(a+h) - \Delta^{n-1} f(a)$.

The operator ∇ : Let $y = f(x)$ be any function of x . Let the consecutive values of x be a , $a+h$, $a+2h$,, $a+nh$ differing by h . Then the corresponding values of y are $f(a)$, $f(a+h)$, $f(a+2h)$,, $f(a+nh)$.

The independent variable x known as argument and the dependent variable y is known as entry. Thus we are given a set values of argument and entry.

The difference $f(a) - f(a-h)$ is called the first backward difference of the function $f(x)$ at the point $x=a$ and we denoted by $\nabla f(a)$, i.e.,

$$\nabla f(a) = f(a) - f(a-h)$$

Similarly,

$$\nabla f(a+h) = f(a+h) - f(a)$$

The general formula is

$$\nabla f(a+nh) = f(a+nh) - f(a+(n-1)h)$$

The operator ∇ is called the backward difference operator and the differences $\nabla f(a)$, $\nabla f(a+h)$, $\nabla f(a+2h)$, etc. are called first backward differences.

The difference $\nabla f(a) - \nabla f(a-h)$ is known as the second backward difference of $f(x)$ at the point $x=a$ and is denoted by $\nabla^2 f(a)$, i.e.,

$$\begin{aligned}\nabla^2 f(a) &= \nabla f(a) - \nabla f(a-h) \\ &= \{f(a) - f(a-h)\} - \{f(a-h) - f(a-2h)\} \\ &= f(a) - 2f(a-h) + f(a-2h)\end{aligned}$$

Similarly $\nabla^3 f(a) = \nabla^2 f(a) - \nabla^2 f(a-h)$. In general $\nabla^n f(a) = \nabla^{n-1} f(a) - \nabla^{n-1} f(a-h)$.

Identity operator 1: The operator 1, defined by $1f(x) = f(x)$, is called the identity operator.

Ex 1. Prove that

- (i) $E \equiv 1 + \Delta$ or $\Delta = E - 1$
- (ii) $\nabla \equiv 1 - E^{-1}$ or $E^{-1} \equiv 1 - \nabla$
- (iii) $E\nabla \equiv \nabla E \equiv \Delta$

$$(iv) \quad \Delta - \nabla \equiv \Delta \nabla$$

$$(v) \quad (1 + \Delta)(1 - \nabla) \equiv 1$$

Proof. (i) We have

$$\begin{aligned}\Delta f(x) &= f(x+h) - f(x) \\ &= Ef(x) - f(x) \\ &= (E - 1)f(x)\end{aligned}$$

Thus $\Delta f(x) = (E - 1)f(x)$, for any function $f(x)$. $\therefore \Delta = E - 1$ or $E \equiv 1 + \Delta$.

(ii) We have

$$\begin{aligned}\nabla f(x) &= f(x) - f(x-h) \\ &= f(x) - E^{-1}f(x) \\ &= (1 - E^{-1})f(x)\end{aligned}$$

Thus $\nabla f(x) = (1 - E^{-1})f(x)$, for any function $f(x)$. $\therefore \nabla = 1 - E^{-1}$ or $E^{-1} \equiv 1 - \nabla$.

(iii) We have

$$\begin{aligned}E\nabla f(x) &= E(\nabla f(x)) \\ &= E\{f(x) - f(x-h)\} \\ &= Ef(x) - Ef(x-h) \\ &= f(x+h) - f(x) \\ &= \Delta f(x)\end{aligned}$$

Therefore $E\nabla \equiv \Delta$. Similarly we can show that $\nabla E \equiv \Delta$.

(iv) We get

$$\begin{aligned}\Delta \nabla f(x) &= \Delta\{f(x) - f(x-h)\} \\ &= \Delta f(x) - \Delta f(x-h) \\ &= \Delta f(x) - \{f(x) - f(x-h)\} \\ &= \Delta f(x) - \nabla f(x) \\ &= (\Delta - \nabla)f(x)\end{aligned}$$

Therefore $\Delta \nabla \equiv \Delta - \nabla$.

(v) We have

$$\begin{aligned}(1 + \Delta)(1 - \nabla)f(x) &= (1 + \Delta)\{1f(x) - \nabla f(x)\} \\ &= (1 + \Delta)[f(x) - \{f(x) - f(x-h)\}] \\ &= (1 + \Delta)[f(x) - f(x) + f(x-h)] \\ &= (1 + \Delta)[f(x-h)] \\ &= 1f(x-h) + \Delta f(x-h) \\ &= f(x-h) + f(x) - f(x-h) \\ &= f(x) \\ &= 1f(x)\end{aligned}$$

Therefore $(1 + \Delta)(1 - \nabla) \equiv 1$.

Theorem: Fundamental theorem of difference calculus.

The n th difference of a polynomial of degree n is constant and higher order differences are zero i.e., if $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$ is a polynomial of degree n in x then the n th order difference of $f(x)$ is constant $a_n h^n n!$ and $(n+1)$ th order difference is zero.

Proof. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$, where n is a positive integer and $a_i, i = 0, 1, 2, \dots, n$ are constant.

We have $\Delta f(x) = f(x+h) - f(x)$

$$= a_0 + a_1(x+h) + a_2(x+h)^2 + a_3(x+h)^3 + \dots + a_n(x+h)^n - (a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n)$$

$$= a_1h + a_2[(x+h)^2 - x^2] + a_3[(x+h)^3 - x^3] + \dots + a_n[(x+h)^n - x^n]$$

$$= a_1h + a_2(2C_1xh + h^2) + a_3(3C_1x^2h + 3C_2xh^2 + h^3) + \dots$$

$$+ a_n(nC_1x^{n-1}h + nC_2x^{n-2}h^2 + \dots + nC_nh^n)$$

$$= b_0 + b_1x + b_2x^2 + b_3x^3 + \dots + b_{n-1}x^{n-1} \dots \dots \dots (i)$$

Where $b_0, b_1, b_2, \dots, b_{n-1}$ are constant coefficients. From (i) we see that triangle $f(x)$ is a polynomial of degree $n-1$ in x . Thus the first difference of a polynomial $f(x)$ of degree n is again a polynomial of degree $n-1$ in which the coefficient of $x^{n-1} = n.h$. the coefficient of x^n in $f(x)$

Now let $\Delta f(x) = \phi(x)$ where $\phi(x)$ is a polynomial of degree $n-1$. Then $\Delta^2 f(x) = \Delta\{\Delta f(x)\} = \Delta\phi(x)$. But by (i), $\Delta\phi(x)$ is a polynomial of degree $n-2$ in which the coefficient of $x^{n-2} = (n-1).h$ the coefficient of x^{n-1} in $\phi(x)$

$$= (n-1)h.n.h.a_n$$

$$= n(n-1)h^2a_n$$

Thus $\Delta^2 f(x)$ is a polynomial of degree $n-2$ in which the coefficient of $x^{n-2} = n(n-1)h^2a_n$

Continuing the above process, we see that the n th difference of $f(x)$ is a polynomial of degree zero i.e. $\Delta^n f(x) = n(n-1)(n-2)\dots 1.h^n a_n x^{n-n} = n! h^n a_n$. Thus the n th difference of $f(x)$ is constant. So all higher order differences are zero, i.e. $(n+1)$ th and higher differences of a polynomial of degree n are zero.

Note. The converse of the above result is also true i.e. if the n th differences of a tabulated function are constant when values of the independent variable are taken at equal intervals, the function is a polynomial of degree n .

Ex 2. Show that

$$f(a+nh) = f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + {}^nC_3 \Delta^3 f(a) + \dots + {}^nC_n \Delta^n f(a)$$

Solution. We have

$$\begin{aligned} f(a+nh) &= E^n f(a) \\ &= (1+\Delta)^n f(a) \\ &= (1+{}^nC_1 \Delta + {}^nC_2 \Delta^2 + {}^nC_3 \Delta^3 + \dots + {}^nC_n \Delta^n) f(a) \\ &= f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + {}^nC_3 \Delta^3 f(a) + \dots + {}^nC_n \Delta^n f(a) \end{aligned}$$

(Proved)

Ex 3. Evaluate the following:

(i) $\Delta^3(1-x)(1-2x)(1-3x)$

(ii) $\Delta^n(e^{ax+b})$

The interval of differencing being unity

Solution.

(i) Here $f(x) = (1-x)(1-2x)(1-3x) = 1 - 6x + 11x^2 - 6x^3$, a polynomial of degree 3. We know that The n th difference of a polynomial of degree n is constant being equal to $a_n h^n n!$ where a_n is the coefficient of x^n and h is the interval of differencing. Here $n = 3$, $h = 1$ and $a_n = -6$. So $\Delta^3 f(x) = (-6)(1^3)3! = -36$

(ii) Here $f(x) = e^{ax+b}$

$$\text{Now } \Delta f(x) = f(x+1) - f(x) = e^{a(x+1)+b} - e^{ax+b} = e^{ax+b}(e^a - 1)$$

$$\Delta^2 f(x) = \Delta(\Delta f(x)) = \Delta\{e^{ax+b}(e^a - 1)\} = (e^a - 1)\Delta e^{ax+b}$$

$$= (e^a - 1)(e^a - 1)e^{ax+b}$$

$$= (e^a - 1)^2 e^{ax+b}$$

$$\text{Proceeding in this way we get } \Delta^n f(x) = (e^a - 1)^n e^{ax+b}$$

$$\therefore \Delta^n e^{ax+b} = (e^a - 1)^n e^{ax+b}$$

Ex 4. Evaluate the following:

(i) $\Delta^2(\cos(2x))$

(ii) $\Delta^2(3e^x)$

(iii) $\left(\frac{\Delta^2}{E}\right)x^3$

Solution.

(i) We get

$$\begin{aligned} \Delta^2(\cos(2x)) &= (E-1)^2 \cos(2x) \\ &= (E^2 - 2E + 1) \cos(2x) \\ &= E^2 \cos 2x - 2E \cos 2x + \cos 2x \\ &= \cos\{2(x+2h)\} - 2\cos\{2(x+h)\} + \cos 2x \\ &= \{\cos(2x+4h) - \cos(2x+2h)\} - \{\cos(2x+2h) + \cos 2x\} \\ &= -4 \sin^2 h \cos(2x+2h) \quad (\text{Ans.}) \end{aligned}$$

(ii) We have

$$\begin{aligned}
\Delta^2(3e^x) &= 3\Delta(\Delta e^x) \\
&= 3\Delta(e^{x+h} - e^x) \\
&= 3\{\Delta e^{x+h} - \Delta e^x\} \\
&= 3\{e^{x+2h} - e^{x+h} - (e^{x+h} - e^x)\} \\
&= 3\{e^{x+2h} - e^{x+h} - e^{x+h} + e^x\} \\
&= 3\{e^{x+2h} - 2e^{x+h} + e^x\} \\
&= 3\{e^x e^{2h} - 2e^x e^h + e^x\} \\
&= 3e^x\{e^{2h} - 2e^h + 1\} \\
&= 3e^x\{e^h - 1\}^2 \quad (\text{Ans.})
\end{aligned}$$

(iii) We have $\left(\frac{\Delta^2}{E}\right)x^3 = \left(\frac{(E-1)^2}{E}\right)x^3 = \left(\frac{E^2-2E+1}{E}\right)x^3 = (E-2+E^{-1})x^3 = (x+1)^3 - 2x^3 + (x-1)^3 = 6x$. (Ans.)

Ex 5. Show that $\Delta \ln f(x) = \ln \left\{1 + \frac{\Delta f(x)}{f(x)}\right\}$

Solution. We have

$$\begin{aligned}
\Delta \ln f(x) &= \ln f(x+h) - \ln f(x) \\
&= \ln \frac{f(x+h)}{f(x)} \\
&= \ln \frac{Ef(x)}{f(x)} \\
&= \ln \frac{(1+\Delta)f(x)}{f(x)} \\
&= \ln \frac{f(x) + \Delta f(x)}{f(x)} \\
&= \ln \left[1 + \frac{\Delta f(x)}{f(x)}\right] \quad (\text{Proved})
\end{aligned}$$

Ex 6. Evaluate the following:

- (i) $\Delta \tan^{-1} x$
(ii) $\Delta\{x + \cos x\}$

Solution.

(i) We have

$$\begin{aligned}
\Delta \tan^{-1} x &= \tan^{-1}(x+h) - \tan^{-1} x \\
&= \tan^{-1} \frac{x+h-x}{1+(x+h)x} \\
&= \tan^{-1} \frac{h}{x^2+xh+1} \quad (\text{Ans.})
\end{aligned}$$

(ii) We have

$$\begin{aligned}
\Delta\{x + \cos x\} &= \Delta x + \Delta \cos x \\
&= (x+h) - x + \cos(x+h) - \cos x
\end{aligned}$$

$$\begin{aligned}
&= h + \cos(x+h) - \cos x \\
&= h + 2 \sin \frac{2x+h}{2} \sin \left(-\frac{h}{2}\right) \\
&= h - 2 \sin \frac{2x+h}{2} \sin \left(\frac{h}{2}\right) \quad (\text{Ans.})
\end{aligned}$$

Ex 7. Show that

$$\sum_{k=0}^{n-1} \Delta^2 f_k = \Delta f_n - \Delta f_0$$

Solution.

$$\begin{aligned}
\text{LHS} &= \sum_{k=0}^{n-1} \Delta^2 f_k \\
&= \sum_{k=0}^{n-1} (E-1)^2 f_k \\
&= \sum_{k=0}^{n-1} (E^2 - 2E + 1) f_k \\
&= \sum_{k=0}^{n-1} f_{k+2} - 2f_{k+1} + f_k \\
&= f_2 - 2f_1 + f_0 \\
&\quad + f_3 - 2f_2 + f_1 \\
&\quad + f_4 - 2f_3 + f_2 \\
&\quad + \dots \dots \dots \\
&\quad + f_{n+1} - 2f_n + f_{n-1} \\
&= f_{n+1} - f_n + f_0 - f_1 \\
&= f_{n+1} - f_n - (f_1 - f_0) \\
&= \Delta f_n - \Delta f_0 \\
&= \text{RHS}
\end{aligned}$$

Ex 8. Obtain the missing terms in the following table:

x	1	2	3	4	5	6	7	8
$f(x)$	1	8	?	64	?	216	343	512

Solution. Here $h = 1$ and we are given 6 values of $f(x)$. Therefore a polynomial of 5th degree may be fitted which have its 6th difference is zero.

i.e., for any values of x

$$\begin{aligned}
\Delta^6 f(x) &= 0. \\
\Rightarrow (E-1)^6 f(x) &= 0 \\
\Rightarrow (E^6 - 6E^5 + 15E^4 - 20E^3 + 15E^2 - 6E + 1)f(x) &= 0 \\
\Rightarrow E^6 f(x) - 6E^5 f(x) + 15E^4 f(x) - 20E^3 f(x) + 15E^2 f(x) - 6E f(x) + 1f(x) &= 0 \\
\Rightarrow f(x+6) - 6f(x+5) + 15f(x+4) - 20f(x+3) + 15f(x+2) - 6f(x+1) + f(x) &= 0 \dots \dots \dots (i)
\end{aligned}$$

Putting $x = 1$ and $x = 2$ in (i) we get.

$$\begin{aligned} f(7) - 6f(6) + 15f(5) - 20f(4) + 15f(3) - 6f(2) + f(1) &= 0 \\ \Rightarrow 343 - 6(216) + 15f(5) - 20(64) + 15f(3) - 6(8) + 1 &= 0 \\ \Rightarrow 343 - 6(216) + 15f(5) - 20(64) + 15f(3) - 6(8) + 1 &= 0 \\ \Rightarrow f(5) + f(3) &= 152 \dots \dots \dots (ii) \end{aligned}$$

And

$$\begin{aligned} f(8) - 6f(7) + 15f(6) - 20f(5) + 15f(4) - 6f(3) + f(2) &= 0 \\ \Rightarrow 512 - 6(343) + 15(216) - 20f(5) + 15(64) - 6f(3) + 8 &= 0 \\ \Rightarrow 10f(5) - 3f(3) &= 1331 \dots \dots \dots (iii) \end{aligned}$$

Solving (ii) and (iii) we get

$$f(3) = 27 \text{ and } f(5) = 125 \quad (\text{Ans.})$$

Ex 9. Obtain the missing terms in the following table:

x	0	1	2	3	4
$f(x)$	1	3	9	?	81

Solution. Here the interval of differencing $h = 1$ and we are given 4 values of $f(x)$. Therefore a polynomial of 3rd degree may be fitted which have its 4th difference is zero.

i.e., for any values of x

$$\begin{aligned} \Delta^4 f(x) &= 0. \\ \Rightarrow (E - 1)^4 f(x) &= 0 \\ \Rightarrow (E^4 - 4E^3 + 6E^2 - 4E + 1)f(x) &= 0 \\ \Rightarrow E^4 f(x) - 4E^3 f(x) + 6E^2 f(x) - 4E f(x) + 1f(x) &= 0 \\ \Rightarrow f(x + 4) - 4f(x + 3) + 6f(x + 2) - 4f(x + 1) + f(x) &= 0 \dots \dots \dots (i) \end{aligned}$$

Putting $x = 0$ in (i) we get

$$\begin{aligned} f(4) - 4f(3) + 6f(2) - 4f(1) + f(0) &= 0 \\ \Rightarrow 81 - 4f(3) + 6(9) - 4(3) + 1 &= 0 \\ \Rightarrow f(3) &= 31 \quad (\text{Ans.}) \end{aligned}$$

Ex 10. Find $f(6)$ given that $f(0) = -3$, $f(1) = 6$, $f(2) = 8$, $f(3) = 12$, the third differences being constant.

Solution. We construct the following difference table:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	-3			
1	6	9		
2	8	2	-7	
3	12	4	2	9

$$\begin{aligned} \text{We have } f(6) &= f(0 + 6) \\ &= E^6 f(0) \\ &= (\Delta + 1)^6 f(0) \\ &= (\Delta^6 + 6\Delta^5 + 15\Delta^4 + 20\Delta^3 + 15\Delta^2 + 6\Delta + 1)f(0) \\ &= (20\Delta^3 + 15\Delta^2 + 6\Delta + 1)f(0), \text{ 4th and higher differences are zero} \\ &= (20\Delta^3 f(0) + 15\Delta^2 f(0) + 6\Delta f(0) + 1f(0)) \\ &= 20(9) + 15(-7) + 6(9) - 3 \\ &= 126 \quad (\text{Ans.}) \end{aligned}$$

Ex 11. Find the lowest degree polynomial which takes the following values:

x	0	1	2	3	4	5
$f(x)$	0	3	8	15	24	35

Solution. We construct the following difference table:

x	$f(x)$	Δ	Δ^2	Δ^3
0	0			
1	3	3		
2	8	5	2	
3	15	7	2	0
4	24	9	2	0
5	35	11	2	0

We know that $f(a + nh) = f(a) + {}^nC_1 \Delta f(a) + {}^nC_2 \Delta^2 f(a) + {}^nC_3 \Delta^3 f(a) + \dots \dots \dots$

Put $a = 0, h = 1, n = x$ we get

$$\begin{aligned} f(x) &= 0 + x(3) + \frac{x(x-1)}{2} 2 \\ \Rightarrow f(x) &= 3x + x^2 - x \\ \Rightarrow f(x) &= x^2 + 2x \quad (\text{Ans.}) \end{aligned}$$

Numerical differentiation

Numerical Differentiation

Defn. The process of calculating the derivatives of a function by means of the set of given values of the function is called numerical differentiation.

Differentiation formula by using Newton's forward interpolation formula. Consider Newton's forward difference formula

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 + \dots \quad (1)$$

$$\text{Where } x = x_0 + hp$$

$$p = \frac{x - x_0}{h}$$

$$\frac{dp}{dx} = \frac{1}{h}$$

Now differentiating (1) w. r. to x we get

$$\begin{aligned} y'(x) &= \frac{dy}{dp} \cdot \frac{dp}{dx} \\ &= \frac{dp}{dx} \cdot \frac{dy}{dp} \\ &= \frac{1}{h} (\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \frac{4p^3+18p^2+22p-6}{4!}\Delta^4 y_0 + \dots) \end{aligned} \quad (2)$$

Again differentiating (2) w.r.to x we get,

$$y'' = \frac{1}{h^2} (\Delta^2 y_0 + \frac{6p-6}{3!}\Delta^3 y_0 + \frac{12p^2+36p+22}{4!}\Delta^4 y_0 + \dots) \quad (3)$$

Formula (2) and (3) can be used for computing the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for non-tabular values of x . The formula takes a simpler form by using $x=x_0$ we obtain, $p=0$, Hence from (2), we get

$$\begin{aligned} [y']_{x=x_0} &= \frac{1}{h} [\Delta y_0 + \frac{-1}{2!}\Delta^2 y_0 + \frac{2}{3!}\Delta^3 y_0 + \frac{-6}{4!}\Delta^4 y_0 + \dots] \\ \text{Or, } [y']_{x=x_0} &= \frac{1}{h} [\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \dots] \end{aligned} \quad (4)$$

Again from (3) we obtain,

$$\begin{aligned} [y'']_{x=x_0} &= \frac{1}{h^2} [\Delta^2 y_0 + \frac{-6}{3!}\Delta^3 y_0 + \frac{22}{4!}\Delta^4 y_0 + \dots] \\ \text{Or, } [y'']_{x=x_0} &= \frac{1}{h^2} [\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \dots] \end{aligned} \quad (5)$$

Equation (2) and (4) gives the 1st derivative and (3) and (5) give the 2nd derivative. Similarly formula for higher derivatives may be obtained by successive differentiation.

Ex 1. Find the 1st and 2nd derivatives of the function tabulated below at the point $x=3$.

x	3.0	3.2	3.4	3.6	3.8	4.0
$y=f(x)$	-14	-10.032	-5.296	.256	6.672	14

Soln. The difference table is as below.

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5
3.0	-14.00					
		3.968				
3.2	-10.032		.768			
		4.736		.048		
3.4	-5.296		.816		0	
		5.552		.048		0
3.6	.256		.864		0	
		6.416		.048		
3.8	6.672		.912			
		7.328				
4.0	14.00					

We know the differentiation formulas by using the Newton's forward formula are

$$y' = \frac{1}{h} [\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots] \quad (1)$$

$$y'' = \frac{1}{h^2} [\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \dots] \quad (2)$$

Here, $x_0 = 3.0$, $h = .2$, $y_0 = -14.00$

$\Delta y_0 = 3.968$, $\Delta^2 y_0 = .768$, $\Delta^3 y_0 = .048$, $\Delta^4 y_0 = 0$

Hence from (1) we get,

$$[y']_{x=3.0} = \frac{1}{.2} \left[3.968 - \frac{1}{2} (.768) + \frac{1}{3} (.048) \right]$$

$$= 18$$

And from (2) we get,

$$[y'']_{x=3.0} = \frac{1}{(.2)^2} [.786 - .048 + 0]$$

$$= 18$$

Ex 2. Find the 1st and 2nd derivatives of the function tabulated below at the point $x=1.1$

Soln. Since the derivatives are required at $x=1.1$, which is near the beginning of the table, we shall use Newton's Forward formula.

The difference formula is as below

x	y	Δ	Δ^2	Δ^3	Δ^4
1.0	0				
		.1280			
1.2	.1280		.280		
		.4160		.0480	
1.4	.5440		.3360		0
		.7520		.0480	
1.6	1.2960		.3840		0
		1.1360		.0480	
1.8	2.4320		.4320		
		1.5680			
2.0	4.00				

We know Newton's Forward difference formula is

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$$

$$+ \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots \quad (1)$$

$$\text{Where } p = \frac{x-x_0}{h}$$

$$\frac{dp}{dx} = \frac{1}{h}$$

Differentiating (1) w.r.to x we get,

$$y'(x) = \frac{1}{h} [\Delta y_0 + \frac{2p-1}{2} \Delta^2 y_0 + \frac{3p^2-6p+2}{6} \Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{24} \Delta^4 y_0 + \dots] \quad (2)$$

Again differentiating (2) w. r. to x

$$y''(x) = \frac{1}{h^2} [\Delta^2 y_0 + \frac{6p-6}{6} \Delta^3 y_0 + \frac{12p^2-36p+22}{24} \Delta^4 y_0 + \dots] \quad (3)$$

Here, $x_0 = 1.0$, $h = .2$, $x = 1.1$

$$p = \frac{x-x_0}{h} = \frac{1.1-1.0}{.2} = \frac{.1}{.2} = .5$$

Hence from (2) we get,

$$y'(1.1) = \frac{1}{.2} [.1280 + \frac{2(.5)-1}{2} (.2880) + \frac{3(.5)^2-6(.5)+2}{6} (.0480)]$$

$$= .630$$

Again from (3) we get,

$$y''(1.1) = \frac{1}{(.2)^2} [.2880 + \frac{6(.5)-6}{6} (.0480)] = 0.660$$

Derivative using Newton's backward difference formula. Consider Newton's backward formula:

$$y(x) = y_n + k \nabla y_n + \frac{k(k+1)}{2!} \nabla^2 y_n + \frac{k(k+1)(k+2)}{3!} \nabla^3 y_n + \frac{k(k+1)(k+2)(k+3)}{4!} \nabla^4 y_n + \dots \quad (1)$$

$$\text{Where } x = x_n + kh, \quad k = \frac{x - x_n}{h}$$

$$\frac{dk}{dx} = \frac{1}{h}$$

Now differentiating (1) w.r.to x we get

$$y'(x) = \frac{dy}{dx} \cdot \frac{dk}{dx} = \frac{1}{h} \frac{dy}{dk} \\ = \frac{1}{h} [\nabla y_n + \left(k + \frac{1}{2}\right) \nabla^2 y_n + \left(\frac{k^2}{2} + k + \frac{1}{3}\right) \nabla^3 y_n + \left(\frac{k^3}{6} + \frac{3}{4}k^2 + \frac{11}{12}k + \frac{1}{4}\right) \nabla^4 y_n + \dots] \quad (2)$$

Again differentiating (2) w. r. to x

$$y''(x) = \frac{1}{h^2} [\nabla^2 y_n + (k+1) \nabla^3 y_n + \left(\frac{1}{2}k^2 + \frac{3}{2}k + \frac{11}{12}\right) \nabla^4 y_n + \dots] \quad (3)$$

Formula (2) and (3) can be used for computing the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ respectively for non-tabular value of x .

For tabular value of x , the formula takes a simpler form for by using $x = x_n$ i.e. $k = 0$.

Then (2) and (3) becomes

$$f'(x_n) = \frac{1}{h} [\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots] \quad (4)$$

$$f''(x_n) = \frac{1}{h^2} [\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \dots] \quad (5)$$

(4) and (5) gives 1st and 2nd derivative.

Ex 3. Find $f'(2.2)$ and $f''(2.2)$ from the following table.

x	1.4	1.6	1.8	2.0	2.2
$f(x)$	4.0552	4.9530	6.0496	7.3891	9.0250

Soln. The difference table is as below:

x	$y = f(x)$	∇	∇^2	∇^3	∇^4
1.4	4.0552	.8918			
1.6	4.9530	1.0966	.1988		
1.8	6.0496	1.3395	.2429	.0441	
2.0	7.3891	1.6359	.2964	.0535	.0094
2.2	9.0250				

Here $x_n = 2.2$, $y_n = 9.0250$, $\nabla f_n = 1.6359$, $\nabla^2 y_n = .2964$, $\nabla^3 y_n = .0535$, $\nabla^4 y_n = .0094$, $h = .02$

We know

$$f'(x_n) = \frac{1}{h} [\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots]$$

$$f'(2.2) = \frac{1}{.02} \left[1.6359 + \frac{1}{2} (.2264) + \frac{1}{3} (.0535) + \frac{1}{4} (.0094) \right] = 9.021416667$$

Again

$$f''(x_n) = \frac{1}{h^2} [\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \dots]$$

$$f''(2.2) = \frac{1}{(.02)^2} \left[.2964 + .0535 + \frac{11}{12} (.0094) \right] = 8.962916667$$

Derivative based on Stirling's formula:

We know the Stirling's formula is

$$f(x) = f_0 + p \frac{\Delta f_0 + \Delta f_{-1}}{2} + \frac{p^2}{2} \Delta^2 f_{-1} + \frac{p(p^2-1)}{3!} \frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \frac{p^2(p^2-1)}{4!} \Delta^4 f_{-2} + \dots \quad (1)$$

$$\text{Where } p = \frac{x - x_0}{h} \text{ i.e. } \frac{dp}{dx} = \frac{1}{h}$$

Differentiation (1) w. r. to x , we get

$$f'(x) = \frac{1}{h} \left[\frac{\Delta f_0 + \Delta f_{-1}}{2} + p \Delta^2 f_{-1} + \frac{3p^2-1}{3!} \frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \frac{4p^3-2p}{4!} \Delta^4 f_{-2} + \dots \right] \quad (2)$$

Again differentiating (2) w. r. to x

$$f''(x) = \frac{1}{h^2} \left[\Delta^2 f_{-1} + p \frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \frac{12p^2-2}{4!} \Delta^4 f_{-2} + \dots \right] \quad (3)$$

For the point $x = x_0$ we get $p = 0$. Thus (2) and (3) becomes

$$f'(x_0) = \frac{1}{h} \left[\frac{\Delta f_0 + \Delta f_{-1}}{2} - \frac{1}{6} \frac{\Delta^3 f_{-1} + \Delta^3 f_{-2}}{2} + \dots \right] \quad (4)$$

$$f''(x_0) = \frac{1}{h^2} [\Delta^2 f_{-1} - \frac{1}{2} \Delta^4 f_{-2} + \dots] \quad (5)$$

(2) and (4) gives the 1st derivative and (3) and (5) gives the 2nd derivative.

Ex 4. Given the table

x	0	1	2	3	4
$f(x)$	6.9897	7.4036	7.7815	8.1281	8.4510

Find $f'(2)$ and $f''(2)$ using appropriate formula.

Soln. Here we use the derivative which is obtained from Stirling's formula.

The table is

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4
0	6.9897				
		.4139			
1	7.4036		-.0366		
		.3779		.0047	
2	7.7815		-.0313		.0025
		.3466		.0076	
3	8.1281		-.0237		
		.3224			
4	8.4510				

Here $x_0 = 2$, $h = 1$

$$f_0 = 7.7815, \quad \Delta f_0 = .3466, \quad \Delta f_{-1} = .3779,$$

$$\Delta^2 f_{-1} = -.0313, \quad \Delta^3 f_{-1} = .0076, \quad \Delta^3 f_{-2} = .0047, \quad \Delta^4 f_{-2} = .0025$$

We know that

$$f'(x_0) = \frac{1}{h} \left[\frac{\Delta f_0 + \Delta f_{-1}}{2} - \frac{1}{3!} \frac{\Delta^3 f_{-2} + \Delta^3 f_{-1}}{2} + \dots \right]$$

$$f'(2) = \frac{1}{1} \left[\frac{.3779 + .3466}{2} - \frac{1}{6} \frac{.0047 + .0076}{2} \right] = .361225$$

$$\text{And } f''(x_0) = \frac{1}{h^2} \left[\Delta^2 f_{-1} - \frac{1}{12} \Delta^4 f_{-2} + \dots \right]$$

$$f''(2) = \frac{1}{1^2} [-.0313 - \frac{1}{2} (.0025)] = -.03151$$

Ex: Given the data

x	0	1	2	4
$f(x)$	5	14	41	98

Use the Lagrange's interpolation formula to find $f'(0.9)$.

Soln: We know the Lagrange's formula is

$$\begin{aligned} f(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f(x_0) + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f(x_1) \\ &\quad + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f(x_2) + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f(x_3) \\ &= \frac{(x-1)(x-2)(x-4)}{(0-1)(0-2)(0-4)} (5) + \frac{(x-0)(x-2)(x-4)}{(1-0)(1-2)(1-4)} (14) \\ &\quad + \frac{(x-0)(x-1)(x-4)}{(2-0)(2-1)(2-4)} (41) + \frac{(x-0)(x-1)(x-2)}{(4-0)(4-1)(4-2)} (98) \\ &= \frac{x^3 - 7x^2 + 14x - 8}{-8} (5) + \frac{x^3 - 6x^2 + 8x}{3} (14) \\ &\quad + \frac{x^3 - 5x^2 + 4x}{-4} (41) + \frac{x^3 - 3x^2 + 2x}{24} (98) \\ &= \left(-\frac{5}{8} + \frac{14}{3} - \frac{41}{4} + \frac{98}{24} \right) x^3 + \left(\frac{35}{8} - 28 + \frac{205}{4} - \frac{3}{24} \right) x^2 + \\ &\quad \left(-\frac{35}{4} + \frac{112}{3} - 41 + \frac{98}{12} \right) x + 5 \\ &\therefore f(x) = -\frac{17}{8} x^3 + \frac{123}{8} x^2 - \frac{17}{4} x + 5 \\ f'(x) &= \frac{-51}{8} x^2 + \frac{123}{4} x - \frac{17}{4} \\ f'(.9) &= \frac{-51}{8} (.9)^2 + \frac{123}{4} (.9) - \frac{17}{4} \\ &= 18.26125 \end{aligned}$$

Ex 5. Use the following data to find $f'(5)$

x	0	2	3	4	7	9
$f(x)$	4	26	58	112	466	922

Soln. Here we use Newton’s divided difference formula. The table is

x	$f(x)$	1 st divided difference	2 nd divided difference	3 rd divided difference	4 th divided difference
0	4				
		11			
2	26		7		
		32		1	
3	58		11		0
		54		1	
4	112		16		0
		118		1	
7	466		22		
		228			
9	922				

We know the Newton’s divided difference formula is

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + \dots$$

We can take $x_0 = 0, x_1 = 2, x_2 = 3, x_3 = 4$

Therefore $f(x_0) = 4, f(x_0, x_1) = 11, f(x_0, x_1, x_2) = 7, f(x_0, x_1, x_2, x_3) = 1$

$$\begin{aligned} \therefore f(x) &= 4 + (x - 0)11 + (x - 0)(x - 2)7 + (x - 0)(x - 2)(x - 3)1 \\ &= 4 + 11x + 7x^2 - 14x + x^3 - 5x^2 + 6x \\ &= x^3 + 2x^2 + 3x + 4 \end{aligned}$$

So

$$\begin{aligned} f'(x) &= 3x^2 + 4x + 3 \\ f'(5) &= 3(5)^2 + 4(5) + 3 \\ &= 98 \end{aligned}$$

Numerical integration

A General Quadrature Formula for Equidistance Ordinates. Let a set of data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of a function $y = f(x)$ be given, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y dx \quad (1)$$

Let the interval $[a, b]$ be divided into n equal subintervals each length h such that $a = x_0 < x_1 < x_2 < \dots < x_n = b$. Clearly, $x_n = x_0 + nh$. Hence (1) becomes

$$I = \int_{x_0}^{x_n} y dx$$

Approximating y by Newton's forward difference interpolation formula, we obtain

$$I = \int_{x_0}^{x_n} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dx$$

Since $x = x_0 + ph$

$$dx = h dp$$

$$x = x_0 \Rightarrow p = 0$$

$$x = x_n \Rightarrow p = n \text{ and hence the above integral becomes}$$

$$I = h \int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dp$$

Which gives on simplification

$$I = h \left[ny_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{6} + \dots \right] \quad (2)$$

This is the general formula.

Trapezoidal Rule. Let a set of data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of a function $y = f(x)$ be given, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y dx \quad (1)$$

Let the interval $[a, b]$ be divided into n equal subintervals each length h such that $a = x_0 < x_1 < x_2 < \dots < x_n = b$. Clearly, $x_n = x_0 + nh$. Hence (1) becomes

$$I = \int_{x_0}^{x_n} y dx$$

Approximating y by Newton's forward difference interpolation formula, we obtain

$$I = \int_{x_0}^{x_p} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dx$$

Since $x = x_0 + ph$

$$dx = hdp$$

$$x = x_0 \Rightarrow p = 0$$

$$x = x_n \Rightarrow p = n \text{ and hence the above integral becomes}$$

$$I = h \int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dp$$

Which gives on simplification

$$I = h \left[ny_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{6} + \dots \right] \quad (2)$$

Putting $n=1$ in (2) and neglecting 2nd and higher order differences, we get

$$\int_{x_0}^{x_1} y dx = h \left[y_0 + \frac{1}{2} \Delta y_0 \right] = h \left[y_0 + \frac{y_1 - y_0}{2} \right] = h \left[\frac{y_0 + y_1}{2} \right]$$

Similarly $\int_{x_1}^{x_2} y dx = h \left[\frac{y_1 + y_2}{2} \right]$

$$\int_{x_2}^{x_3} y dx = h \left[\frac{y_2 + y_3}{2} \right]$$

.....

$$\int_{x_{n-1}}^{x_n} y dx = h \left[\frac{y_{n-1} + y_n}{2} \right]$$

Adding those n integrals, we get

$$\int_{x_0}^{x_n} y dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n]$$

Which is known as trapezoidal rule.

Ex. 1 Calculate the value of the integral $\int_4^{5.2} \ln x dx$ by Trapezoidal rule.

Soln. Divide the interval [4, 5.2] into six equal parts. So that $h=(5.2-4)/6=0.2$ and the value of $\ln x$ for each point of sub division are given below:

n	x	y=ln x
0	4.0	1.3862944
1	4.2	1.4350845
2	4.4	1.4816045

3	4.6	1.5260563
4	4.8	1.5686159
5	5.0	1.6094379
6	5.2	1.6486586

We have from the trapezoidal rule

$$\begin{aligned} \int_{x_0}^{x_6} y dx &= \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + y_4 + y_5) + y_6] \\ \therefore \int_4^{5.2} \ln x dx &= \frac{0.2}{2} [1.3862944 + 2(1.4350845 + 1.4816045 \\ &\quad + 1.5260563 + 1.5686159 + 1.6094379) + 1.6486586] \\ &= 0.1(18.276551) \\ &= 1.8276551 \end{aligned}$$

Ex 2. A curve is drawn to pass through the points given by the following table:

x	1	1.5	2	2.5	3	3.5	4
y	2	2.4	2.7	2.8	3	2.6	2.1

Find the area bounded by the curve, the x-axis and the lines $x=1$, $x=4$

Soln. In order to find the required area we shall compute the value of the integral

$$I = \int_1^4 y dx$$

Here $n=6$, therefore $h=(4-1)/6=0.5$

We have from the trapezoidal rule

$$\begin{aligned} \int_{x_0}^{x_6} y dx &= \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + y_4 + y_5) + y_6] \\ \therefore \int_1^4 y dx &= \frac{0.5}{2} [2 + 2(2.4 + 2.7 + 2.8 + 3 + 2.6) + 2.1] = 7.775 \end{aligned}$$

Simpson's 1/3 Rule. Let a set of data points (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , (x_n, y_n) of a function $y = f(x)$ be given, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y dx \quad (1)$$

Let the interval [a,b] be divided into n equal subintervals each length h such that

$a = x_0 < x_1 < x_2 < \dots < x_n = b$. Clearly, $x_n = x_0 + nh$. Hence (1) becomes

$$I = \int_{x_0}^{x_n} y dx$$

Approximating y by Newton's forward difference interpolation formula, we obtain

$$I = \int_{x_0}^{x_n} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dx$$

Since $x = x_0 + ph$

$$dx = hdp$$

$$x = x_0 \Rightarrow p = 0$$

$x = x_n \Rightarrow p = n$ and hence the above integral becomes

$$I = h \int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dp$$

Which gives on simplification

$$I = h \left[ny_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{6} + \dots \right] \quad (2)$$

Putting $n=2$ in (2) and neglecting 3rd and higher order differences, we get

$$\begin{aligned} \int_{x_0}^{x_2} y dx &= 2h \left[y_0 + \Delta y_0 + \frac{1}{6} \Delta^2 y_0 \right] \\ &= 2h \left[y_0 + (y_1 - y_0) + \frac{1}{6} (y_2 - 2y_1 + y_0) \right] \\ &= \frac{h}{3} [y_0 + 4y_1 + y_2] \end{aligned}$$

Similarly $\int_{x_2}^{x_4} y dx = \frac{h}{3} [y_2 + 4y_3 + y_4]$

$$\int_{x_4}^{x_6} y dx = \frac{h}{3} [y_4 + 4y_5 + y_6]$$

.....

$$\int_{x_{n-2}}^{x_n} y dx = \frac{h}{3} [y_{n-2} + 4y_{n-1} + y_n]$$

Adding those integrals, we get

$$\int_{x_0}^{x_n} y dx = \frac{h}{3} \left[y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n \right]$$

Which is known as Simpson's 1/3 Rule, or simply Simpson's rule. It should be noted that this rule requires the division of the whole range into an even number of subintervals of width h .

Ex 3. Calculate the value of the integral $\int_4^{5.2} \ln x dx$ by Simpson's 1/3 Rule.

Soln. Divide the interval $[4, 5.2]$ into six equal parts. So that $h=(5.2-4)/6=0.2$ and the value of $\ln x$ for each point of sub division are given below:

n	x	$y=\ln x$
0	4.0	1.3862944
1	4.2	1.4350845
2	4.4	1.4816045
3	4.6	1.5260563
4	4.8	1.5686159
5	5.0	1.6094379
6	5.2	1.6486586

We have from Simpson's 1/3 rule

$$\begin{aligned} \int_{x_0}^{x_6} y dx &= \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) + y_6] \\ \therefore \int_4^{5.2} \ln x dx &= \frac{0.2}{3} [1.3862944 + 4(1.4350845 + 1.5260563 + 1.6094379) \\ &\quad + 2(1.4816045 + 1.5686159) + 1.6486586] \\ &= \frac{0.2}{3} (27.417709) = 1.8278472 \end{aligned}$$

Ex 4. Find $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's 1/3 Rule. Hence obtain the approximate value of π .

Soln. Divide the the range of integration $[0, 1]$ into 6 equal parts. So that $h=(1-0)/6=1/6$. the value of $y = \frac{1}{1+x^2}$ for each point of sub division are given below:

n	x	y
0	0	1
1	1/6	0.9729729
2	2/6	0.9
3	3/6	0.8
4	4/6	0.6923076
5	5/6	0.5901639
6	1	0.5

By Simpson's 1/3 Rule we get

$$\int_{x_0}^{x_6} y dx = \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) + y_6]$$

$$\therefore \int_0^1 \frac{dx}{1+x^2}$$

$$= \frac{1/6}{3} \left[1 + 4(0.9729729 + 0.8 + 0.5901639) \right]$$

$$= \frac{1}{18} (14.137163)$$

$$= 0.7853979$$

$$\text{But } \int_0^1 \frac{dx}{1+x^2} = \left[\tan^{-1} x \right]_0^1 = \tan^{-1} 1 - \tan^{-1} 0 = \pi/4$$

Therefore $\pi/4 = 0.7853979$

$$\therefore \pi = 4(0.7853979) = 3.1415916$$

Ex 5. Evaluate $\int_{0.5}^{0.7} x^{1/2} e^{-x} dx$ by using Simpson's 1/3 Rule.

Soln. Divide the range of integration [0.5, 0.7] into 4 equal parts. So that $h=(0.7-0.5)/4=0.05$. the value of $y = x^{1/2} e^{-x}$ for each point of sub division are given below:

n	x	y
0	0.50	0.4288818
1	0.55	0.4278774
2	0.60	0.4251076
3	0.65	0.4208867
4	0.70	0.4154730

By Simpson's 1/3 Rule we get

$$\int_{x_0}^{x_4} y dx = \frac{h}{3} [y_0 + 4(y_1 + y_3) + 2(y_2) + y_4]$$

$$\therefore \int_0^1 x^{1/2} e^{-x} dx$$

$$= \frac{0.05}{3} \left[0.4288818 + 4(0.4278774 + 0.4208867) \right]$$

$$= 0.0848271$$

Ex 6. Evaluate $\int_{0.2}^{1.4} (\sin x - \ln x + e^x) dx$ by using Simpson's 1/3 Rule.

Soln. Divide the the range of integration [0.2, 1.4] into 12 equal parts. So that $h=(1.4-0.2)/12=0.1$. the value of $y = \sin x - \ln x + e^x$ for each point of sub division are given below:

n	x	y		
0	0.2	3.02950		
1	0.3	2.84935	2.84935	
2	0.4	2.79753		2.79753
3	0.5	2.82130	2.82130	
4	0.6	2.89759		2.89759
5	0.7	3.01464	3.01464	
6	0.8	3.16604		3.16604
7	0.9	3.34829	3.34829	
8	1.0	3.55975		3.55975
9	1.1	3.80007	3.80007	
10	1.2	4.06984		4.06984
11	1.3	4.37050	4.37050	
12	1.4	4.70418		
sum			20.2042	16.4908

By Simpson's 1/3 Rule we get

$$\int_{x_0}^{x_6} y dx = \frac{h}{3} \left[y_0 + 4(y_1 + y_3 + y_5 + y_7 + y_9 + y_{11}) \right]$$

$$+ 2(y_2 + y_4 + y_6 + y_8 + y_{10}) + y_{12}$$

$$\therefore \int_{0.2}^{1.4} (\sin x - \ln x + e^x) dx$$

$$= \frac{0.1}{3} [3.02950 + 4(20.2042) + 2(16.4908) + 4.70418]$$

$$= 4.051059$$

Simpson's 3/8 Rule. Let a set of data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ of a function $y = f(x)$ be given, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y dx \quad (1)$$

Let the interval [a,b] be divided into n equal subintervals each length h such that $a = x_0 < x_1 < x_2 < \dots < x_n = b$. Clearly, $x_n = x_0 + nh$. Hence (1) becomes

$$I = \int_{x_0}^{x_n} y dx$$

Approximating y by Newton's forward difference interpolation formula, we obtain

$$I = \int_{x_0}^{x_n} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dx$$

Since $x = x_0 + ph$

$$dx = hdp$$

$$x = x_0 \Rightarrow p = 0$$

$x = x_n \Rightarrow p = n$ and hence the above integral becomes

$$I = h \int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{6} \Delta^3 y_0 + \dots \right] dp$$

Which gives on simplification

$$I = h \left[ny_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2} + \left(\frac{n^4}{4} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{6} + \dots \right] \quad (2)$$

Putting $n=3$ in (2) and neglecting 4th and higher order differences, we get

$$\begin{aligned} \int_{x_0}^{x_3} y dx &= 3h \left[y_0 + \frac{3}{2} \Delta y_0 + \frac{3}{4} \Delta^2 y_0 + \frac{1}{8} \Delta^3 y_0 \right] \\ &= 3h \left[y_0 + \frac{3}{2} (y_1 - y_0) + \frac{3}{4} (y_2 - 2y_1 + y_0) + \frac{1}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right] \end{aligned}$$

$$\int_{x_0}^{x_3} y dx = \frac{3h}{8} [y_0 + 3y_1 + 3y_2 + y_3]$$

Similarly $\int_{x_3}^{x_6} y dx = \frac{3h}{8} [y_3 + 3y_4 + 3y_5 + y_6]$

$$\int_{x_6}^{x_9} y dx = \frac{3h}{8} [y_6 + 3y_7 + 3y_8 + y_9]$$

.....

$$\int_{x_{n-3}}^{x_n} y dx = \frac{3h}{8} [y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n]$$

Adding those integrals, we get

$$\int_{x_0}^{x_n} y dx = \frac{3h}{8} \left[y_0 + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-2} + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3}) + y_n \right]$$

Which is known as Simpson's 3/8 Rule.

Ex 7. Calculate the value of the integral $\int_4^{5.2} \ln x dx$ by Simpson's 3/8 Rule.

Soln. Divide the interval $[4, 5.2]$ into six equal parts. So that $h=(5.2-4)/6=0.2$ and the value of $\ln x$ for each point of sub division are given below:

n	x	$y=\ln x$
0	4.0	1.3862944
1	4.2	1.4350845
2	4.4	1.4816045
3	4.6	1.5260563
4	4.8	1.5686159
5	5.0	1.6094379
6	5.2	1.6486586

We have from Simpson's 3/8 rule

$$\begin{aligned} \int_{x_0}^{x_6} y dx &= \frac{3h}{8} [y_0 + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) + y_6] \\ \therefore \int_4^{5.2} \ln x dx &= \frac{3 \times 0.2}{8} \left[1.3862944 + 3(1.4350845 + 1.4816045 + 1.5686159 + 1.6094379) + 2(1.5260563) + 1.6486586 \right] \\ &= \frac{0.6}{8} (24.371294) = 1.827847 \end{aligned}$$

Ex 8. Calculate the value of the integral $\int_0^1 e^{-x^2} dx$ by Simpson's 3/8 Rule.

Soln. Divide the interval $[0, 1]$ into 10 equal parts. So that $h=(1-0)/10=0.1$ the value of e^{-x^2} for each point of sub division are given below:

n	x	$y=e^{-x^2}$
0	0.0	1.00000
1	0.1	0.99005
2	0.2	0.96079
3	0.3	0.91393
4	0.4	0.85214
5	0.5	0.77880
6	0.6	0.69768
7	0.7	0.61263
8	0.8	0.52729
9	0.9	0.44486
10	1.0	0.36788

We have from Simpson's 3/8 rule

$$\int_{x_0}^{x_{10}} y \, dx = \frac{3h}{8} \left[y_0 + 3(y_1 + y_2 + y_4 + y_5 + y_7 + y_8) + 2(y_3 + y_6 + y_9) + y_{10} \right]$$

$$\therefore \int_0^1 e^{-x^2} \, dx$$

$$= \frac{3 \times 0.1}{8} \left[1.00000 + 3(0.99005 + 0.96079 + 0.85214 + 0.77880 + 0.61263 + 0.52729) + 2(0.91393 + 0.69768 + 0.44486) + 0.36788 \right]$$

$$= \frac{0.3}{8} [1 + 3(4.7222) + 2(2.05647) + 0.36788]$$

$$= 0.73677825$$

System of linear equation

Gausses scidal method. Consider the system of equations

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Now, we solve the 1st equation for x in terms of 'y' and 'z'

$$x = \frac{b_1 - a_{12}y - a_{13}z}{a_{11}} \quad (1)$$

Similarly from the 2nd & 3rd equation

$$y = \frac{b_2 - a_{21}x - a_{23}z}{a_{22}} \quad (2)$$

And

$$z = \frac{b_3 - a_{31}x - a_{32}y}{a_{33}} \quad (3)$$

Next, we make a first approximation to the solution which we call x_1 , y_1 & z_1 (Usually we select zeros for the 1st approximation to the solution).

These 1st approximation are then used in equation (1) to compute a new approximate value for x

$$x_2 = \frac{b_1 - a_{12}y_1 - a_{13}z_1}{a_{11}} \quad (4)$$

$$y_2 = \frac{b_2 - a_{21}x_2 - a_{23}z_1}{a_{22}} \quad (5)$$

$$z_2 = \frac{b_3 - a_{31}x_2 - a_{32}y_2}{a_{33}} \quad (6)$$

Proceeding in this way we get the general equation for the Gausses scidal iterations are

$$x_{n+1} = \frac{b_1 - a_{12}y_n - a_{13}z_n}{a_{11}}$$

$$y_{n+1} = \frac{b_2 - a_{21}x_{n+1} - a_{23}z_n}{a_{22}}$$

$$z_{n+1} = \frac{b_3 - a_{31}x_{n+1} - a_{32}y_{n+1}}{a_{33}}$$

Ex 1. Solve the system by Gausses scidal method.

$$2x + y = 4$$

$$x - y = -1$$

Soln. Our initial estimation of the solution

$$x_1 = 0, y_1 = 0$$

$$\text{Iteration 1: } x_2 = \frac{4 - y_1}{2} = \frac{4 - 0}{2} = 2$$

$$y_2 = \frac{-1 - x_2}{-1} = \frac{-1 - 2}{-1} = 3$$

$$\text{Iteration 2: } x_3 = \frac{4 - y_2}{2} = \frac{4 - 3}{2} = 0.5$$

$$y_3 = \frac{-1 - x_3}{-1} = \frac{-1 - 0.5}{-1} = 1.5$$

$$\text{Iteration 3: } x_4 = \frac{4 - y_3}{2} = \frac{4 - 1.5}{2} = 1.25$$

$$y_4 = \frac{-1 - x_4}{-1} = \frac{-1 - 1.25}{-1} = 2.25$$

Thus the approximate solution is $x = 1.25$ & $y = 2.25$

Ex 2. Solve the system

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

Soln. Solving the equations for the unknown we can rewrite the equations

$$x = \frac{1}{27}(85 - 6y + z) \quad (1)$$

$$y = \frac{1}{15}(72 - 6x - 2z) \quad (2)$$

$$z = \frac{1}{54}(110 - x - y) \quad (3)$$

Our initial estimations are

$$x_0 = 0, y_0 = 0 \text{ \& } z_0 = 0$$

1st iteration:

$$x_1 = \frac{1}{27}(85 - 6y_0 + z_0) = \frac{85}{27} = 3.15$$

$$y_1 = \frac{1}{15}(72 - 6x_1 - 2z_0) = \frac{1}{15}\{72 - (6 \times 3.15)\} = 3.54$$

$$z_1 = \frac{1}{54}(110 - y_1 - x_1) = 1.91$$

$$2^{\text{nd}} \text{ Iteration: } x_2 = \frac{1}{27}(85 - 6y_1 + z_1) = \frac{1}{27}\{85 - (6 \times 3.54) + 1.91\} = 2.43$$

$$y_2 = \frac{1}{15}(72 - 6x_2 - 2z_1) = \frac{1}{15}\{72 - (6 \times 2.43) - (2 \times 1.91)\} = 3.57$$

$$z_2 = \frac{1}{54}(110 - y_2 - x_2) = \frac{1}{54}(110 - 3.57 - 2.43) = 1.93$$

3rd iteration:

$$x_3 = \frac{1}{27}(85 - 6y_2 + z_2) = \frac{1}{27}\{85 - (6 \times 3.57) + 1.93\} = 2.43$$

$$y_3 = \frac{1}{15}(72 - 6x_3 - 2z_2) = \frac{1}{15}\{72 - (6 \times 2.43) - (2 \times 1.93)\} = 3.57$$

$$z_3 = \frac{1}{54}(110 - y_3 - x_2) = \frac{1}{54}(110 - 3.57 - 2.43) = 1.93$$

Thus the approximate result is $x = 2.43, y = 3.57 \text{ \& } z = 1.93$

Gauss elimination method (Forward elimination and backward substitution).

Consider the system

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = c_1 \quad (1)$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = c_2 \quad (2)$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = c_n \quad (3)$$

The first phase is designed to reduce the set of equation to an upper triangular system.

The initial step in the process in to divide equation (1) (pivot equation) by the constant a_{11} , (the pivot coefficient)

$$x_1 + \frac{a_{12}}{a_{11}}x_2 + \dots + \frac{a_{1n}}{a_{11}}x_n = \frac{c_1}{a_{11}} \quad (4)$$

The above process is called normalizations.

Next multiply the normalized equation (4) by the first coefficient of second equation, we get

$$a_{21}x_1 + (a_{21}\frac{a_{12}}{a_{11}})x_2 + \dots + (a_{21}\frac{a_{1n}}{a_{11}})x_n = a_{21}\frac{c_1}{a_{11}} \quad (5)$$

The 1st unknown can be eliminated from the 2nd equation by substituting equation (5) from (2)

$$(a_{22} - a_{21}\frac{a_{12}}{a_{11}})x_2 + \dots + (a_{2n} - a_{21}\frac{a_{1n}}{a_{11}})x_n = c_2 - a_{21}\frac{c_1}{a_{11}}$$

$$\Rightarrow a'_{22}x_2 + \dots + a'_{2n}x_n = c'_2$$

Repeating the process for the remaining equation, the result become in the following modified system.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = c_1 \quad (6)$$

$$a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n = c'_2 \quad (7)$$

$$a_{32}x_2 + a_{33}x_3 + \dots + a_{3n}x_n = c'_3 \quad (8)$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$a'_{n2}x_2 + a'_{n3}x_3 + \dots + a'_{nn}x_n = c'_n \quad (9)$$

Now repeat the above process in order to eliminate the 2nd unknown from equation (8) through (9), we get

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = c_1$$

$$a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n = c'_2$$

$$a''_{33}x_3 + \dots + a''_{3n}x_n = c''_3$$

$$a''_{n3}x_3 + \dots + a''_{nn}x_n = c''_n$$

Where the double prime indicates that the eliminates have been changed or modified twice

The procedure can be continued using the remaining pivot equations, we get

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = c_1 \quad (10)$$

$$a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n = c'_2 \quad (11)$$

$$a''_{33}x_3 + \dots + a''_{3n}x_n = c''_3 \quad (12)$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$a_{nn}^{(n-1)}x_n = c_n^{(n-1)} \quad (13)$$

$$\text{From (13), we get } x_n = \frac{c_n^{(n-1)}}{a_{nn}^{(n-1)}} \quad (14)$$

Remaining x 's can be represented by the following formula

$$x_i = \frac{c_i^{(i-1)} - \sum_{j=i+1}^n a_{ij}^{(i-1)}x_j}{a_{ii}^{(i-1)}} \quad (15)$$

[where $i = n-1, n-2, \dots, 3, 2, 1$]

Ex 3. Use Gauss elimination method to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.3x_3 + 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

Carry six significant figures during computation.

Soln. Now normalized the equation (1) by dividing the pivot coefficient 3 we get

$$x_1 - 0.033333x_2 - 0.066666x_3 = 2.61667 \quad (4)$$

Next, multiplying equation (4) by 0.1 and subtracting the result from (2) we get

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (5)$$

Now to eliminate x_1 multiplying equation (4) by 0.3 and subtracting the result from (3) we get the new system

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (6)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (7)$$

$$-0.190000x_2 + 10.0200x_3 = 70.6156 \quad (8)$$

Now normalized the equation (7) dividing by 7.00333 we get

$$x_2 - 0.0418848x_3 = -2.79320 \quad (9)$$

Then multiplying the equation (9) by -0.190000 and subtracting the result from (8)

$$3x_1 - 0.1x_2 - 0.3x_3 = 7.85$$

$$7.00333x_2 - 0.293333x_3 = -19.5617$$

$$10.0120x_3 = 70.0843$$

We can now solve the equations by backward substitution

$$x_3 = 7.00003$$

$$x_2 = -2.50000$$

$$x_1 = 3.00000$$

Gausses Jordan method. Let the system of equation be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = c_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = c_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = c_3$$

Then the augmented matrix is

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & c_1 \\ a_{21} & a_{22} & a_{23} & c_2 \\ a_{31} & a_{32} & a_{33} & c_3 \end{bmatrix}$$

By any number of row operation the above matrix is reduced to

$$\begin{bmatrix} 1 & 0 & 0 & c'_1 \\ 0 & 1 & 0 & c'_2 \\ 0 & 0 & 1 & c'_3 \end{bmatrix}$$

Thus the solution is $x_1 = c'_1$

$$x_2 = c'_2$$

$$x_3 = c'_3$$

Ex 4. Solve the following system of equation by Gausses Jordan method .

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + 4z = 1$$

Soln. The augmented matrix of the above system is

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & 4 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 3 & -5 \end{bmatrix} \quad [R'_2 = R_2 - R_1], [R'_3 = R_3 - R_1]$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -9 \end{bmatrix} \quad [R'_3 = R_3 - R_2]$$

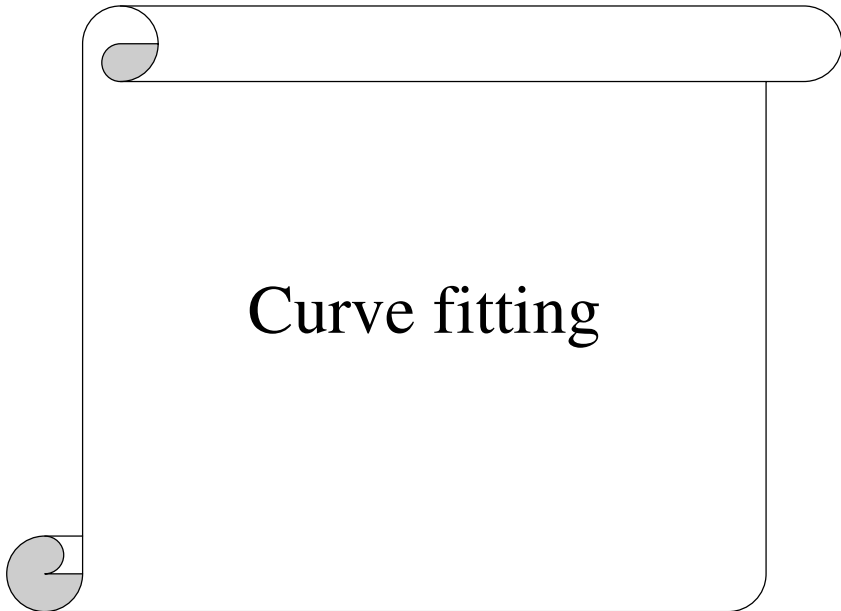
$$\sim \begin{bmatrix} 1 & 1 & 0 & 15 \\ 0 & 1 & 0 & 22 \\ 0 & 0 & 1 & -9 \end{bmatrix} \quad [R'_1 = R_1 - R_3], [R'_2 = R_2 - 2R_3]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 22 \\ 0 & 0 & 1 & -9 \end{bmatrix} \quad [R'_1 = R_1 - R_2]$$

Thus the solutions are $x = -7$

$$y = 22$$

$$z = -9$$



Curve fitting

Curve Fitting

Suppose we have m observations $(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$ of two variables x and y and it is requested to fit a curve of the type

$$y = a + bx + cx^2 + \dots + kx^n \quad \dots \dots \dots (1)$$

Now we have to determine the constants $a, b, \dots, k$ such that it represents the curve of best fit of that degree. In case $m=n$, we get in general a unique set of values satisfying the given systems of equations. But if $m>n$, then by substituting the different values of x and y in equation (1) we get m equations and we want to find only n constants. Hence there may be no such solution to satisfy all m equations. So we try to obtain those values of $a, b, \dots, k$ which may give the best fit i.e. which may satisfy all the equations as nearly as possible.

In such a case the principle of least squares asserts a suitable method.

Putting $x_1, x_2, \dots, x_m$ for x in equation (1), we have

$$Y_1 = a + bx_1 + cx_1^2 + \dots + kx_1^n$$

$$Y_2 = a + bx_2 + cx_2^2 + \dots + kx_2^n$$

$$\dots \dots \dots$$

$$Y_m = a + bx_m + cx_m^2 + \dots + kx_m^n$$

The quantities $Y_1, Y_2, \dots, Y_m$ are the expected values of y corresponding to $x = x_1, x = x_2, \dots, x = x_m$ respectively. The values $y_1, y_2, \dots, y_m$ are the observed values of y corresponding to the given values $x_1, x_2, \dots, x_m$ of x . In general the expected values are different from the observed values.

Let $R_r = y_r - Y_r$. For different values of r these differences are called residuals.

The quantity $R_1^2 + R_2^2 + \dots + R_m^2$ provides a measure of the 'goodness of fit' of the curve to the given data.

If it is small, the fit is good, if it is large the fit is bad.

Now set

$$\begin{aligned} U &= \sum R_i^2 \\ &= \sum (y_r - Y_r)^2 \\ &= \sum (y_r - a - bx_r - cx_r^2 - \dots)^2 \end{aligned}$$

By this principle of least square the constants $a, b, c, \dots, k$ are chosen in such a manner that sum of square of residuals in minimum. For maximum or minimum of U , we must have

$$\frac{\partial U}{\partial a} = 0 = \frac{\partial U}{\partial b} = \frac{\partial U}{\partial c} = \dots = \frac{\partial U}{\partial k}$$

After simplifying these relations, we have

$$\begin{aligned} \sum y &= ma + b \sum x + \dots + k \sum x^n \\ \sum xy &= a \sum x + b \sum x^2 + \dots + k \sum x^{n+1} \\ \sum x^2 y &= a \sum x^2 + b \sum x^3 + \dots + k \sum x^{n+2} \\ &\dots \dots \dots \\ \sum x^n y &= a \sum x^n + b \sum x^{n+1} + \dots + k \sum x^{2n} \end{aligned}$$

These equations are $(n+1)$ in number and can be solved as simultaneous equations to give the values of the constants $a, b, \dots, k$. These equations are called normal equations.

If we calculate the second order partial derivatives and the values $a, b, \dots, k$ are put in these derivatives. They give a positive value of the function.

So U is minimum.

Particular case.

If $n = 1$, then the curve to be fitted is a straight line $y = a + bx$ and the normal equations are

$$\begin{aligned}\sum y &= ma + b\sum x \\ \sum xy &= a\sum x + b\sum x^2\end{aligned}$$

If $n = 2$, then the curve to be fitted is a second degree parabola $y = a + bx + cx^2$ and the normal equations are

$$\begin{aligned}\sum y &= ma + b\sum x + c\sum x^2 \\ \sum xy &= a\sum x + b\sum x^2 + c\sum x^3 \\ \sum x^2y &= a\sum x^2 + b\sum x^3 + c\sum x^4\end{aligned}$$

Ex. Fit a straight line to the following data regarding x as the independent variable:

x:	0	1	2	3	4
y:	1	1.8	3.3	4.5	6.3

Solution. Let the straight line to be fitted to the given data be

$y = a + bx$. Then the normal equations are

$$\begin{aligned}\sum y &= ma + b\sum x \\ \sum xy &= a\sum x + b\sum x^2\end{aligned}$$

Now

	x	y	xy	x^2
	0	1	0	0
	1	1.8	1.8	1
	2	3.3	6.6	4
	3	4.5	13.5	9
	4	6.3	25.2	16
Total	10	16.9	47.1	30

In this case $m = 5$, $\sum x = 10$, $\sum y = 16.9$, $\sum xy = 47.1$, $\sum x^2 = 30$

Substituting these values in the normal equations, we have

$$16.9 = 5a + 10b$$

And $47.1 = 10a + 30b$

Solving the above equations, we obtain $a = 0.72$, $b = 1.33$.

Hence the fitted line is

$$y = 0.72 + 1.33x \text{ (answer)}$$

Ex. Fit a second degree parabola to the following data

x:	0	1	2	3	4
y:	1	5	10	22	38

Solution: Let the parabola to be fitted to the given data be $y = a + bx + cx^2$.

$$\begin{aligned}\sum y &= ma + b\sum x + c\sum x^2 \\ \sum xy &= a\sum x + b\sum x^2 + c\sum x^3 \\ \sum x^2 y &= a\sum x^2 + b\sum x^3 + c\sum x^4\end{aligned}$$

Now

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	5	1	1	1	5	5
2	10	4	8	16	20	40
3	22	9	27	81	66	198
4	38	16	64	256	152	608
10	76	30	100	354	243	851

Substituting the values of $\sum x$, $\sum y$ etc in the normal equations, we get

$$76 = 5a + 10b + 30c$$

$$243 = 10a + 30b + 100c$$

$$851 = 30a + 100b + 354c$$

Solving those equations simultaneously, we get

$$a = 1.43$$

$$b = 0.24$$

$$c = 2.21$$

Hence the parabola is

$$y = 1.43 + 0.24x + 2.21x^2 \text{ (answer)}$$

Ex. The weights of a calf taken at weekly intervals are given below. Fit straight line using the method of least squares and calculate the rate of growth per week.

Age(x):	1	2	3	4	5	6	7	8	9	10
Weight (y):	52.2	58.7	65.0	70.2	75.4	81.1	87.2	95.5	102.2	108.4

Solution: Here x and y denote the variables age and weight respectively.

Let the least square line is $y = a + bx$. Then the normal equations are

$$\sum y = ma + b\sum x$$

$$\sum xy = a\sum x + b\sum x^2$$

Now

x	y	x^2	xy
1	52.2	1	52.2
2	58.7	4	117.4
3	65.0	9	195.0
4	70.2	16	280.8
5	75.4	25	377.0
6	81.1	36	486.6
7	87.2	49	610.4
8	95.5	64	764.0
9	102.2	81	919.8
10	108.4	100	1084.0
55	795.9	385	4887.2

Substituting these values in the normal equations, we get

$$795.9 = 10a + 55b$$

$$4887.2 = 55a + 385b$$

Solving above equations, we get

$$a = 45.61$$

$$b = 6.18$$

$\therefore$ The line is

$$y = 45.61 + 6.18x \text{ (answer)}$$

Again we get

$$\frac{dy}{dx} = 6.18$$

$\therefore$ the rate of growth per week is 6.18 units. (Answer)

Ex. Fit an exponential curve of the form $y = ab^x$ to the following data:

x:	1	2	3	4	5	6	7	8
y:	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1

Solution. Here $y = ab^x$

$$\Rightarrow \ln y = \ln a + \ln b \cdot x$$

Let

$$Y = \ln y$$

$$A = \ln a$$

$$B = \ln b$$

$$\therefore Y = A + Bx$$

The normal equations are

$$\sum Y = mA + B\sum x$$

$$\sum xY = A\sum x + B\sum x^2$$

x	y	Y = lny	xY	x ²
1	1.0	0.0000	0.0000	1
2	1.2	0.0792	0.1548	4
3	1.8	0.2553	0.7659	9
4	2.5	0.3979	1.5916	16
5	3.6	0.5563	2.7815	25
6	4.7	0.6721	4.0326	36
7	6.6	0.8195	5.7365	49
8	9.1	0.9590	7.6720	64
36	30.5	3.7393	22.7385	204

Now

∴ From normal equations

$$3.7393 = 8A + 36B$$

$$22.7385 = 36A + 204B$$

Solving these, we get

$$A = 1.8336, B = 0.1406$$

$$\therefore a = e^A = e^{1.8336} = 0.68$$

$$b = e^B = e^{0.1406} = 1.38$$

∴ The required curve is

$$y = ab^x$$

$$y = 0.68 \times 1.38^x$$